

Simple infectious-disease models for planning

WAHA TAKWIMU Short Course

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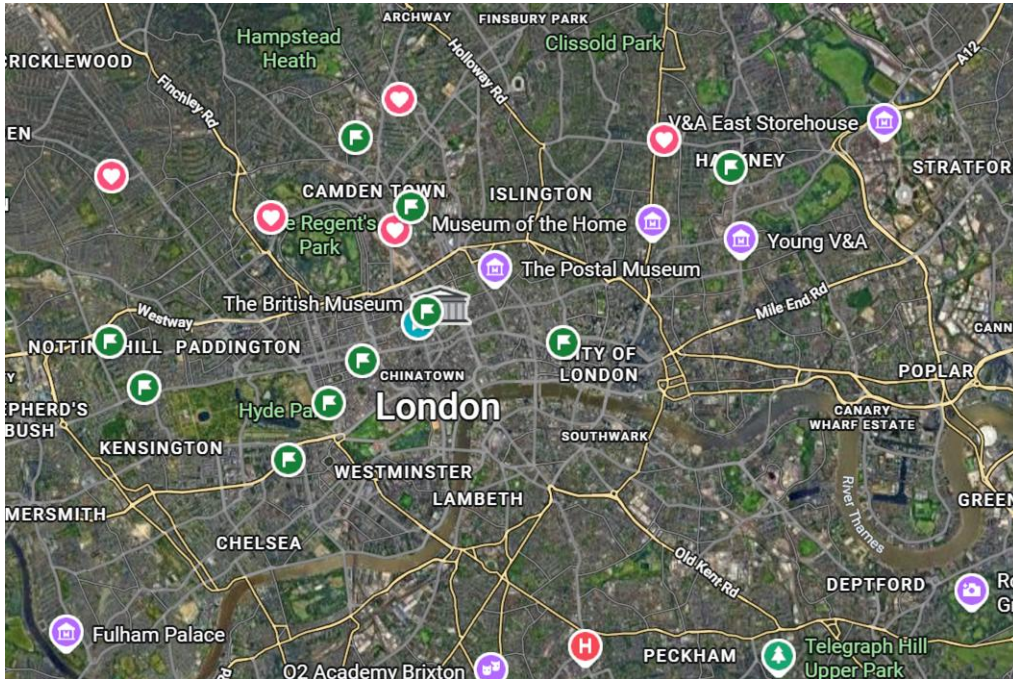
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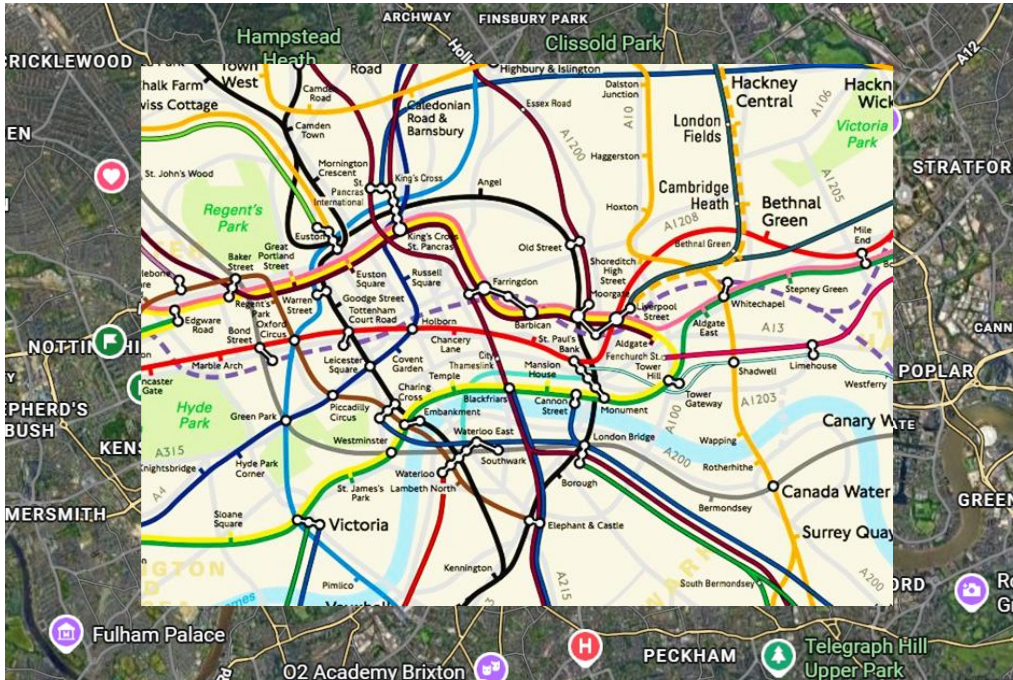
Models simplify reality

They can be used to explore the behavior of a process



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They can be used to explore the behavior of a process

Statistical models

Estimate associations and effect sizes from observed data.

Good for: risk factors, prediction with clear covariates

Limit: weak at feedback loops and counterfactual mechanisms.

Machine learning

Learn high-dimensional patterns with minimal prior structure.

Good for: prediction from large datasets.
Limit: can be difficult to interpret or extrapolate.

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They can be used to explore the behavior of a process

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Mechanistic models

Encode causal processes as flows, rates and feedback loops.

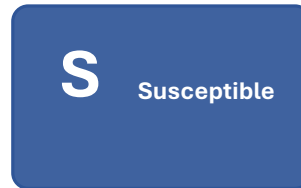
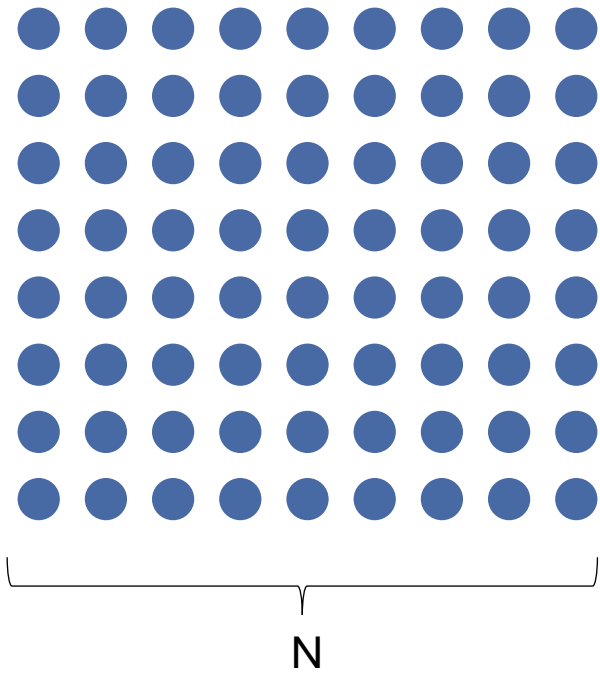
Good for: looped processes, scenarios, interventions, and “what if?” planning.

Limit: assumptions are explicit and must be tested.

→ mechanistic models are transparent reasoning tools: assumptions → equations → consequences

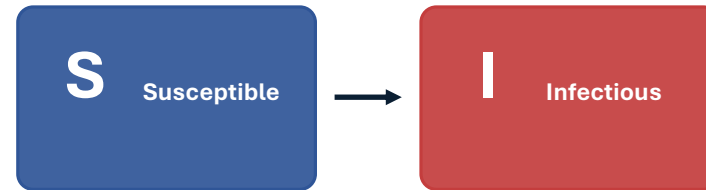
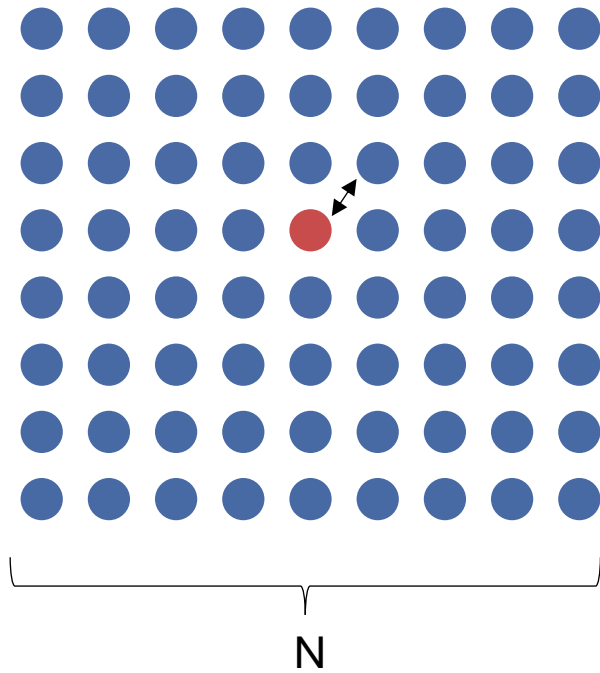
Susceptible, Infectious, Recovered

What do these compartments mean?



Susceptible, Infectious, Recovered

What do these compartments mean?



Closed population assumption

$$S(t) + I(t) = N$$

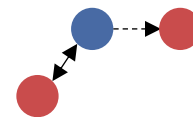
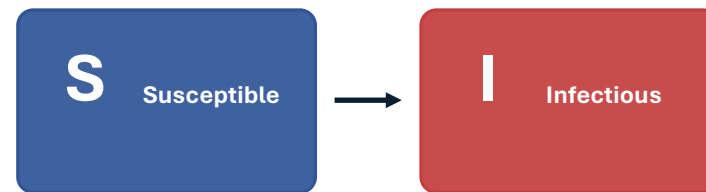
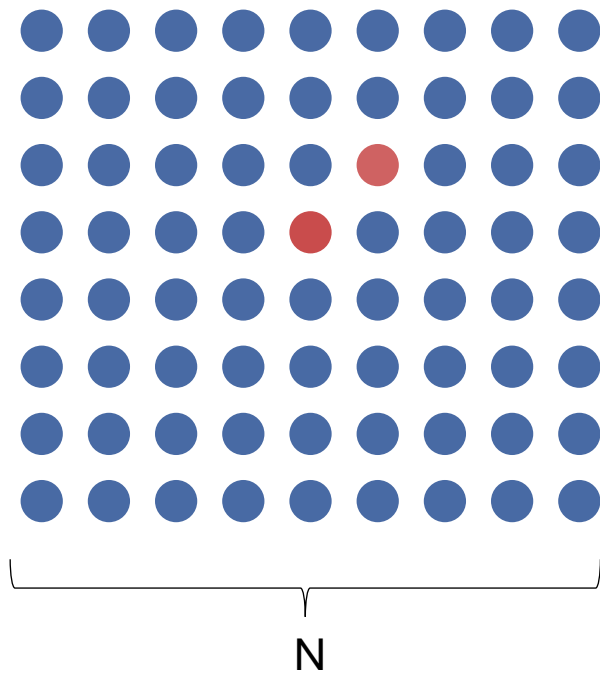
$$S(0) = S_0$$

$$I(0) = I_0$$



Susceptible, Infectious, Recovered

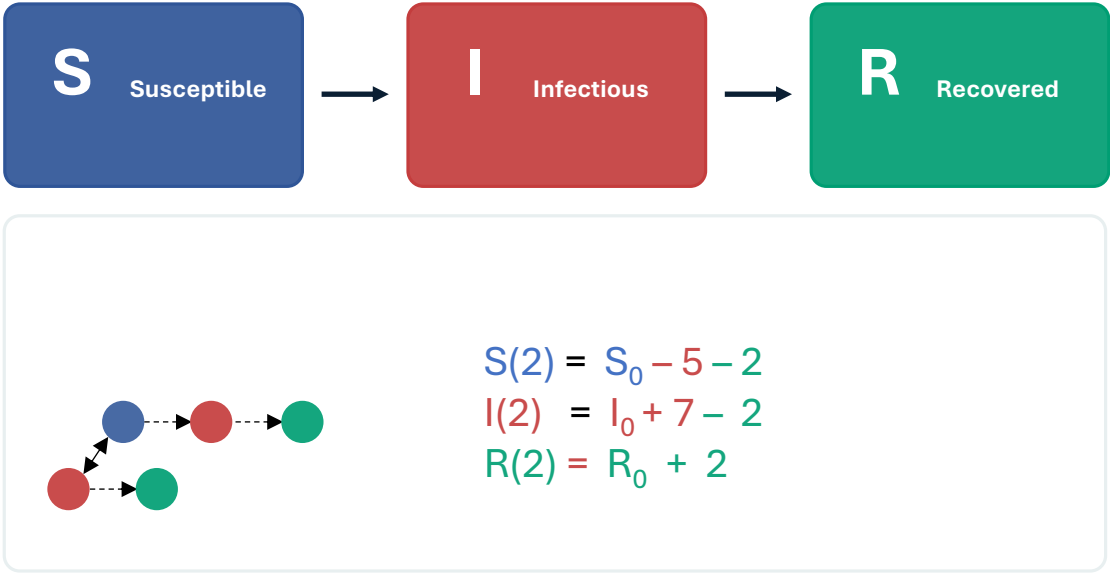
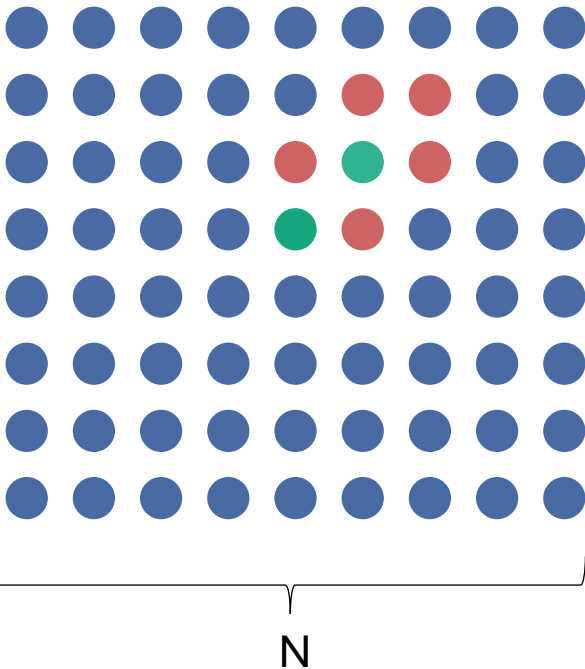
What do these compartments mean?



$$\begin{aligned} S(1) &= S_0 - 2 \\ I(1) &= I_0 + 1 \end{aligned}$$

Susceptible, Infectious, Recovered

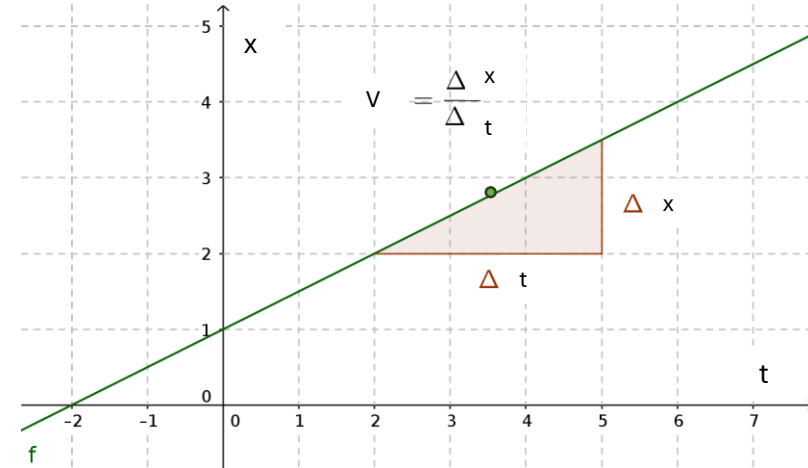
What do these compartments mean?



What are Differential equations?

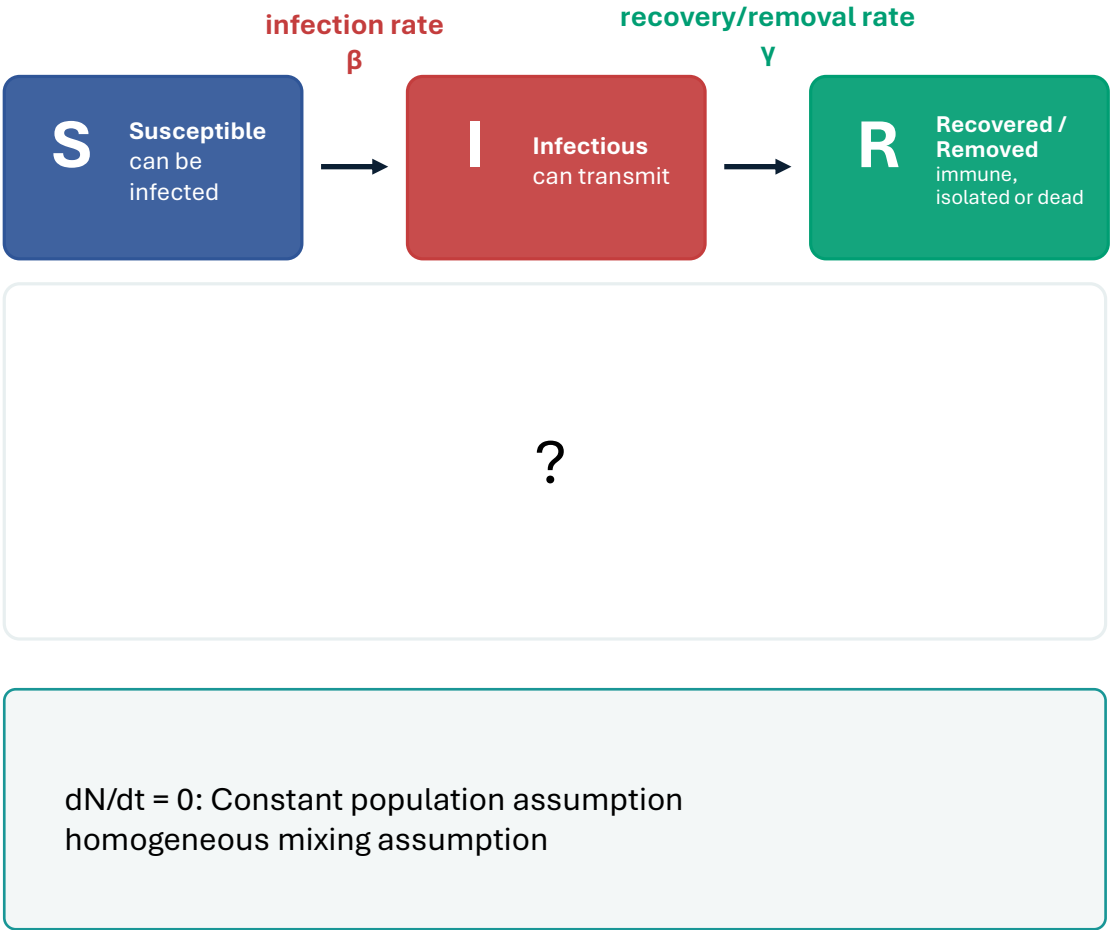
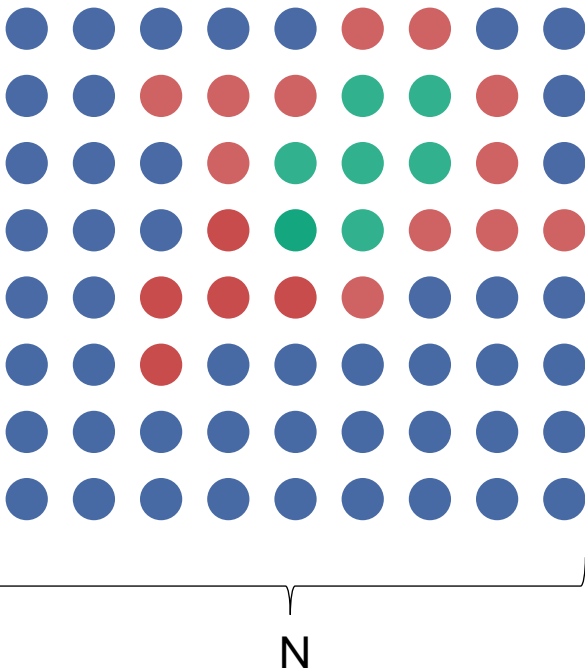
Here to describe change over time – by looking at an infinitesimal small time step

DISPLACEMENT	AVERAGE VELOCITY	AVERAGE ACCELERATION
$\Delta x = x_2 - x_1$	$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$	$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$
	INSTANTANEOUS VELOCITY	INSTANTANEOUS ACCELERATION
	$v = \frac{dx}{dt}$	$a = \frac{dv}{dt}$



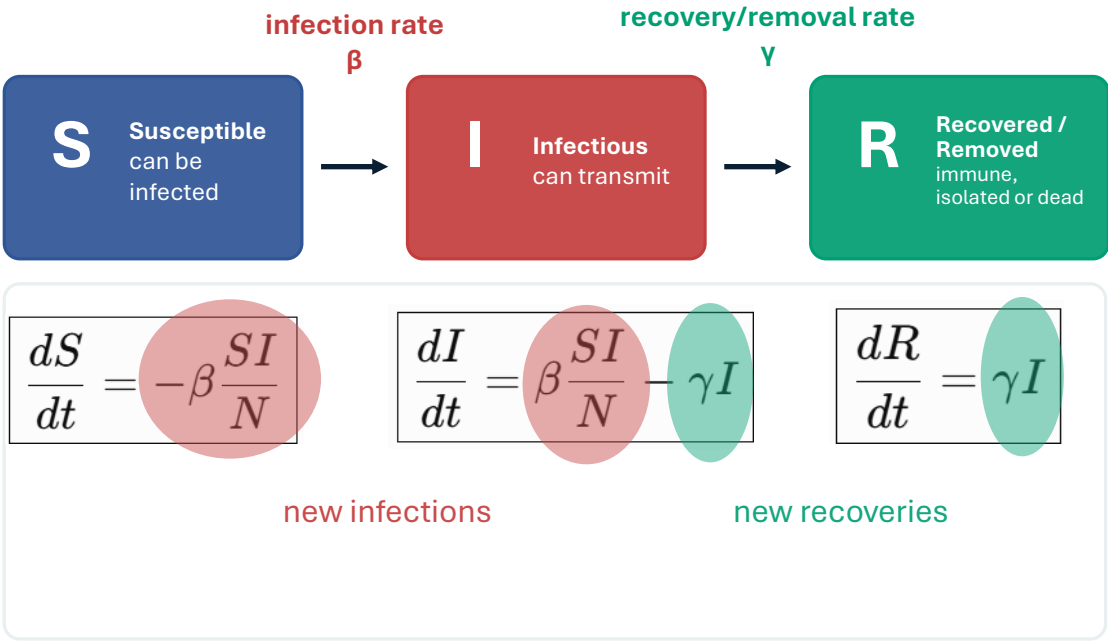
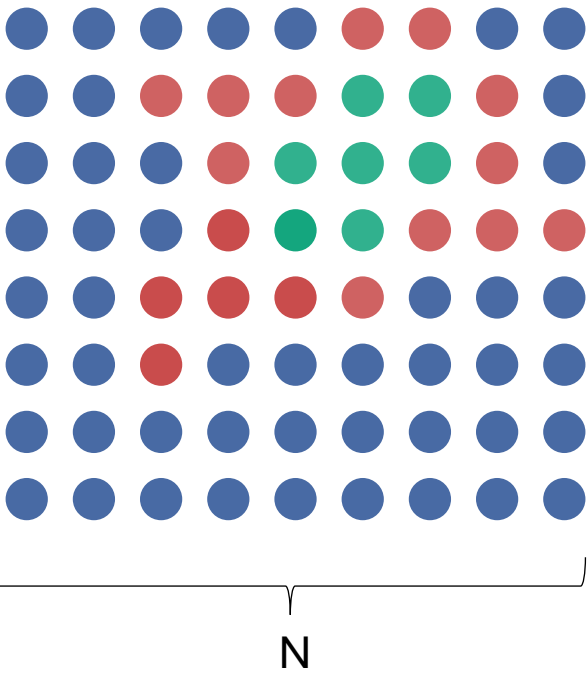
Susceptible, Infectious, Recovered

System of differential equations



Susceptible, Infectious, Recovered

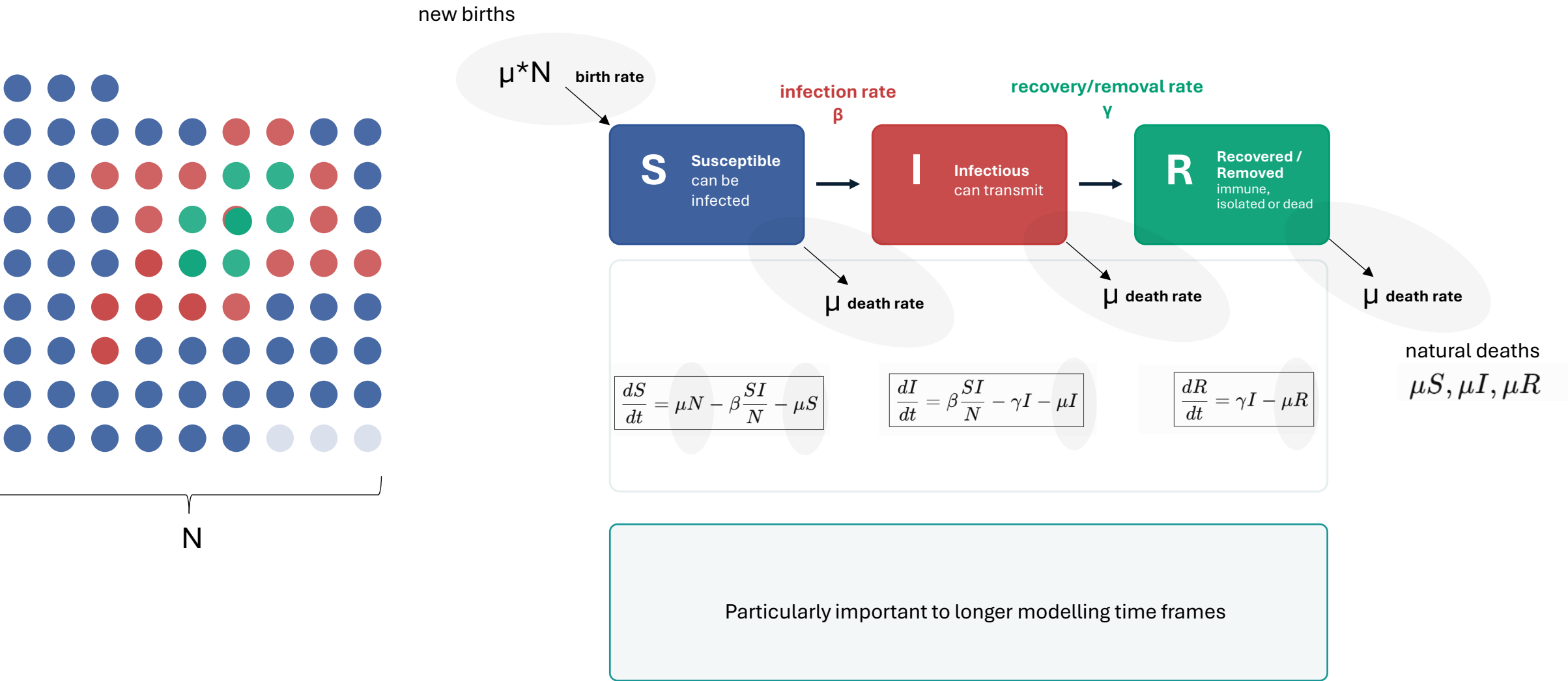
System of differential equations



$S \cdot I$: number of people that could potentially get infected
 $\beta I/N$: force of infection

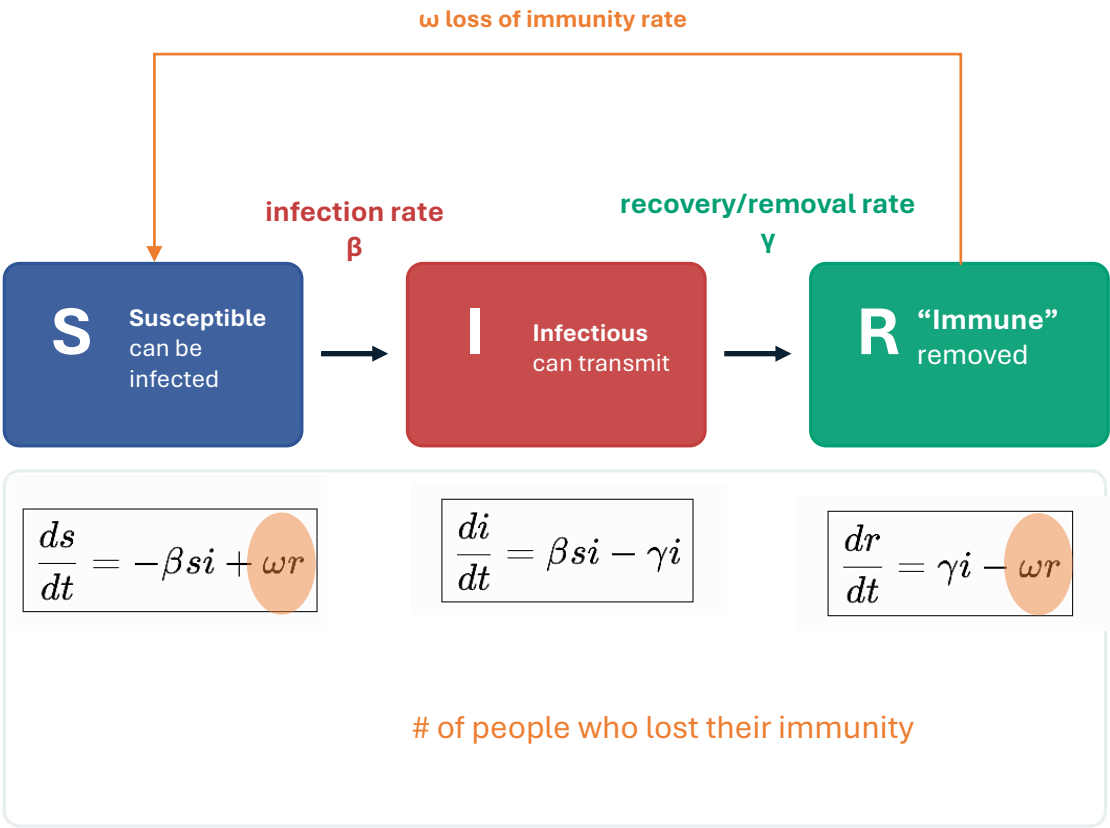
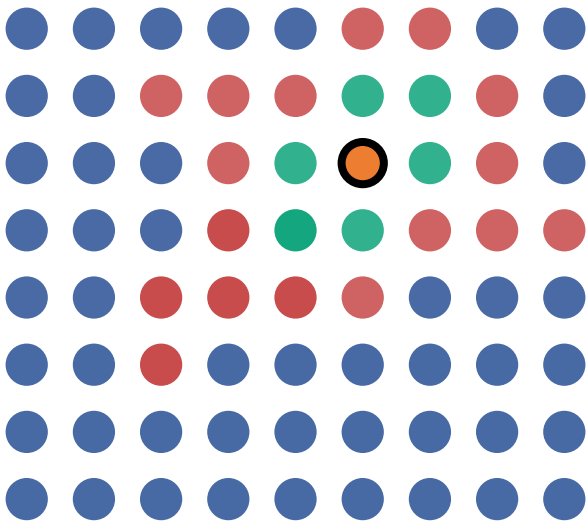
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Susceptible, Infectious, Recovered

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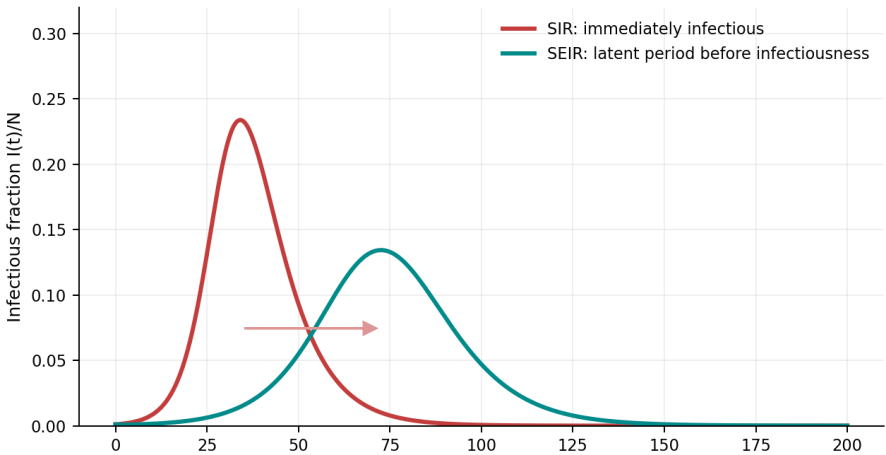
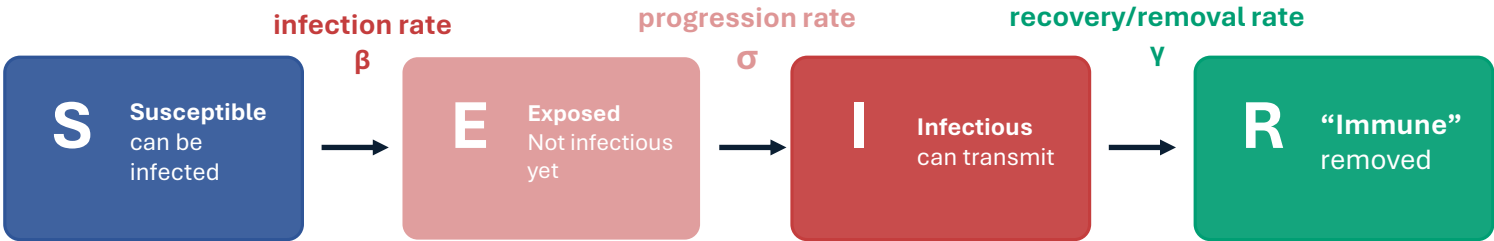
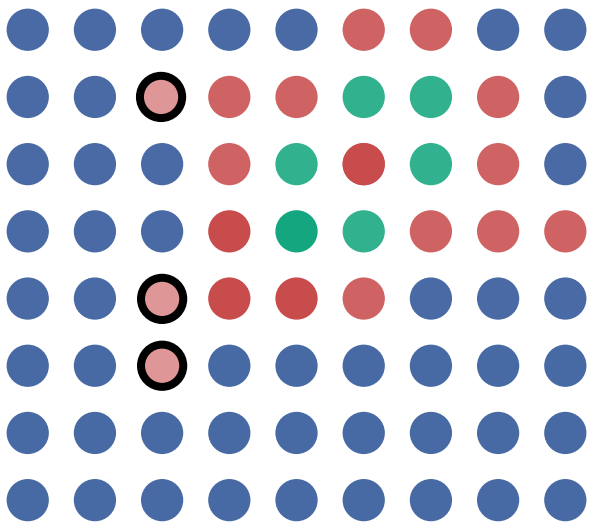


counts: SIR
proportions: sir

$$S + I + R = N$$
$$s = \frac{S}{N}, \quad i = \frac{I}{N}, \quad r = \frac{R}{N},$$
$$s + i + r = 1.$$

Susceptible, Exposed, Infectious, Recovered, Susceptible

SEIRS



$$\frac{ds}{dt} = -\beta si$$
$$\frac{de}{dt} = \beta si - \sigma e$$
$$\frac{di}{dt} = \sigma e - \gamma i$$
$$\frac{dr}{dt} = \gamma i$$

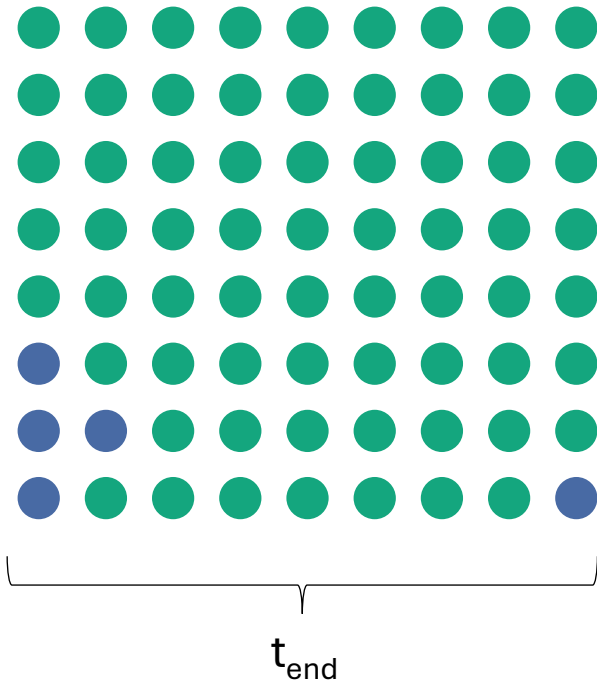
infected people becoming infectious

1/σ: mean latent period

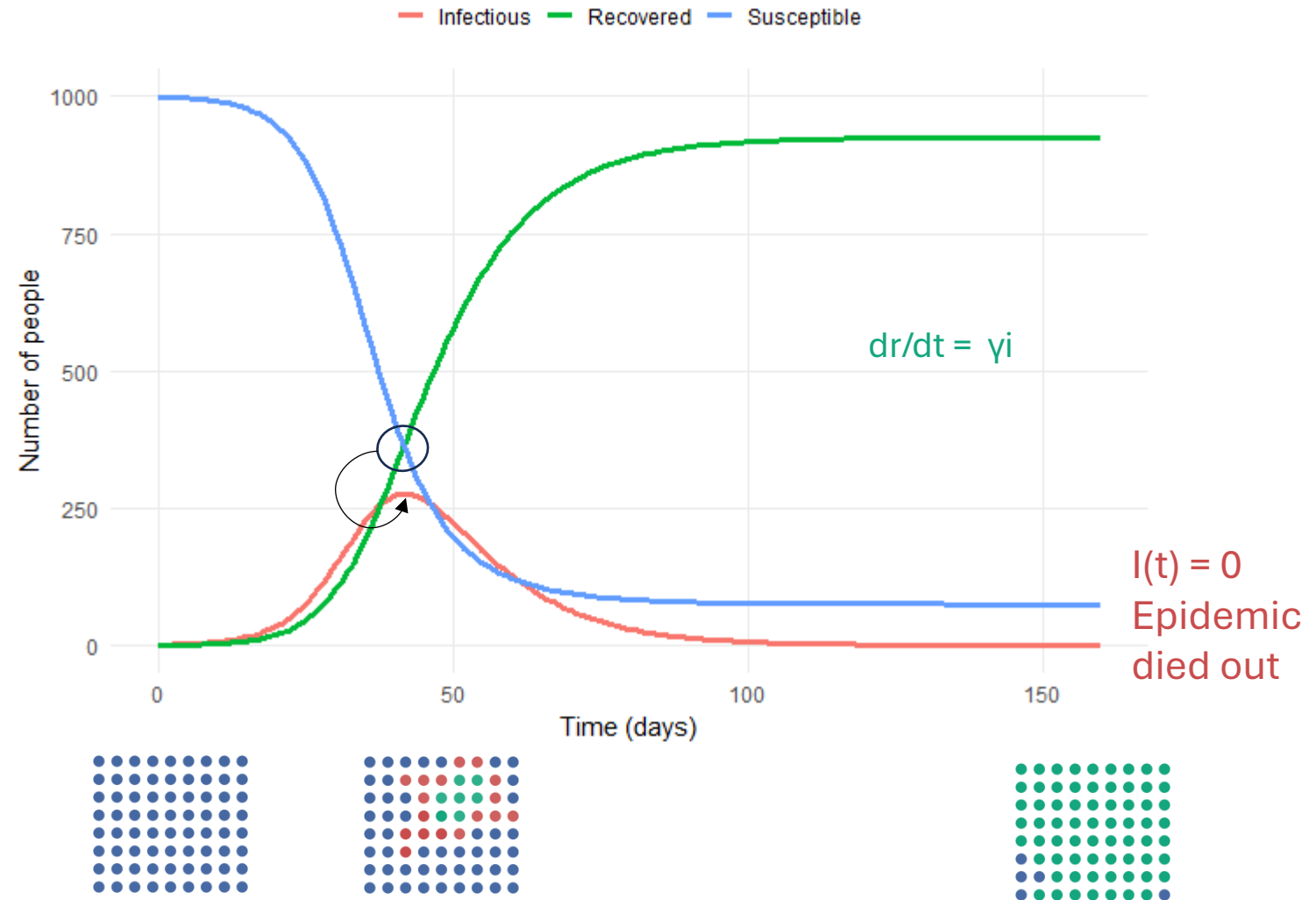
1/γ: mean infectious duration

Dynamics and End

Interpretation of a specific epidemic scenario

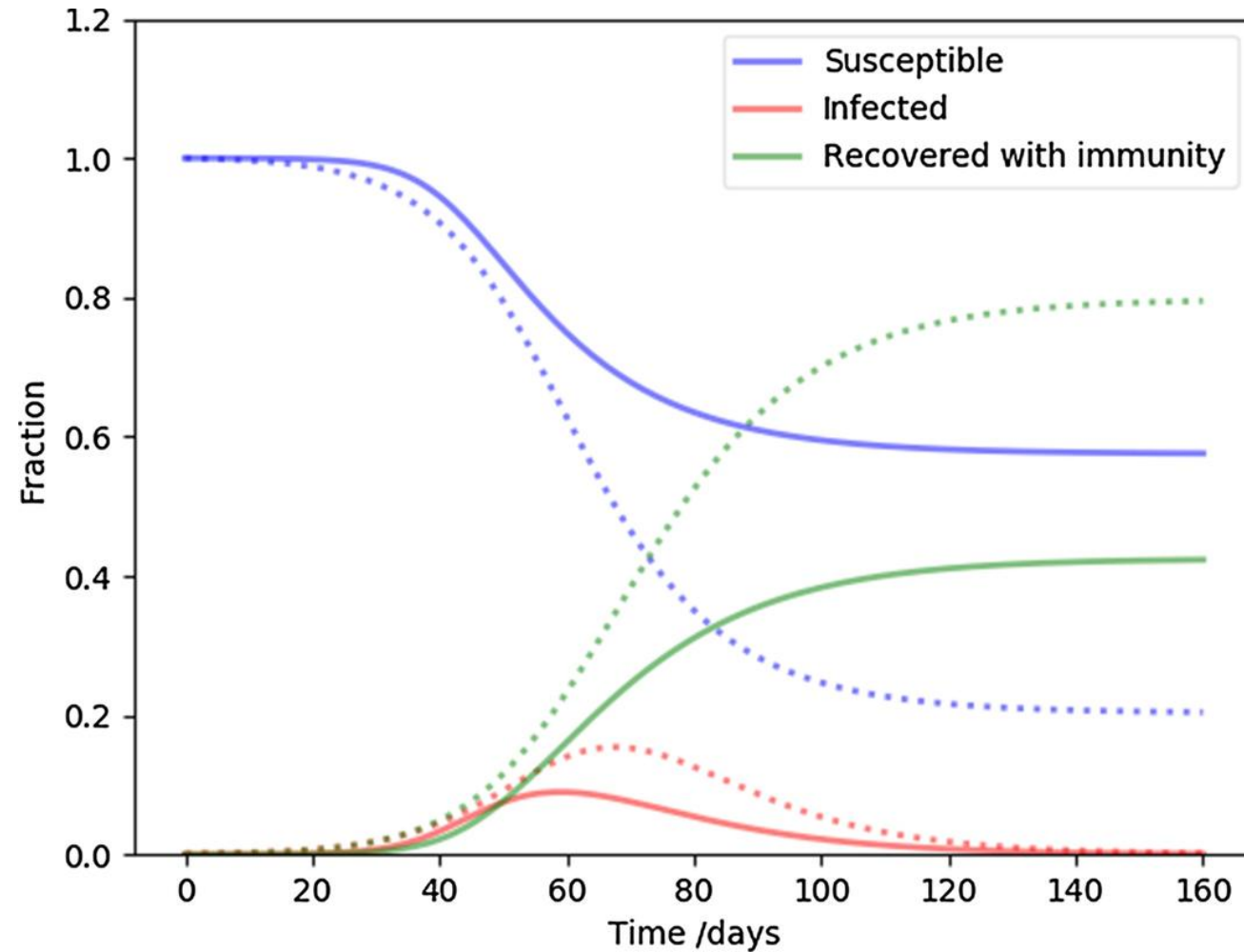
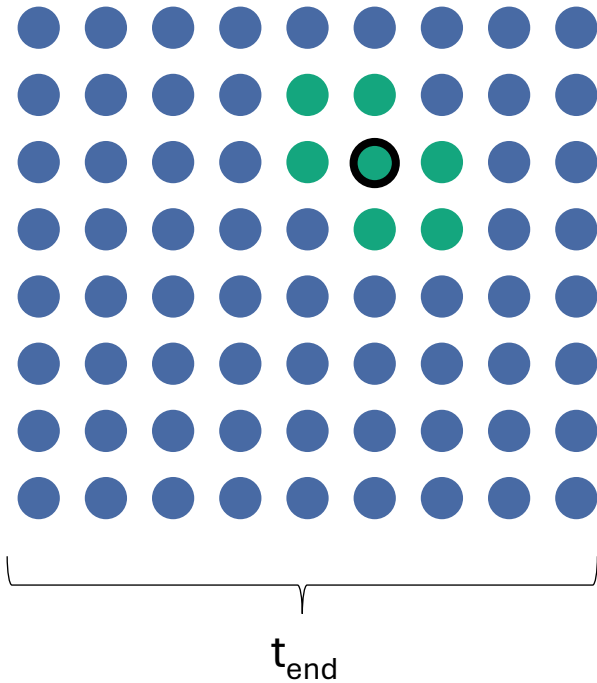


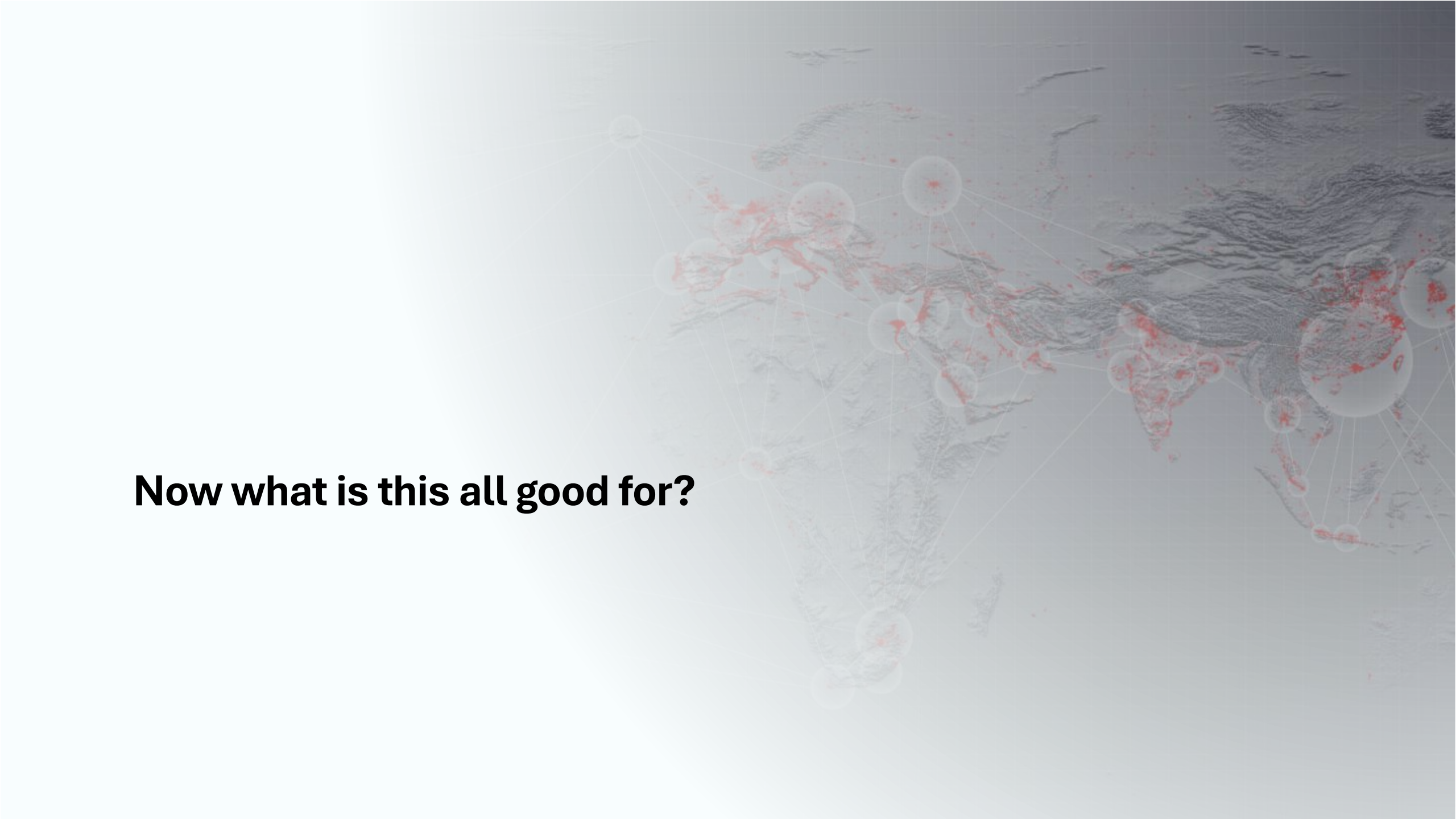
$\beta = 0.28 \text{ day}^{-1}$, $\gamma = 0.1 \text{ day}^{-1}$, $R_0 = 2.8$, $S_0 = 1000$, $I_0 = 1$



Dynamics and End

When the epidemic dies off before reaching everybody it leaves many susceptible

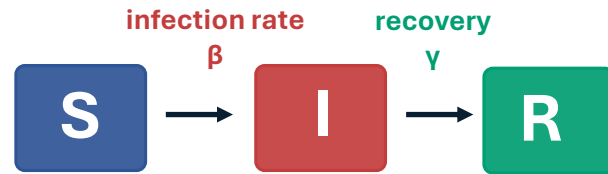


A world map with a network overlay. The map is light gray with a grid. Overlaid on the map is a network of white lines connecting various nodes. The nodes are represented by circles of varying sizes, some of which are filled with red. The red nodes are concentrated in North America, Europe, and Asia, with a few smaller nodes in Africa and South America. The network lines are thin and white, creating a web-like structure across the map.

Now what is this all good for?

Typical characteristics of interest

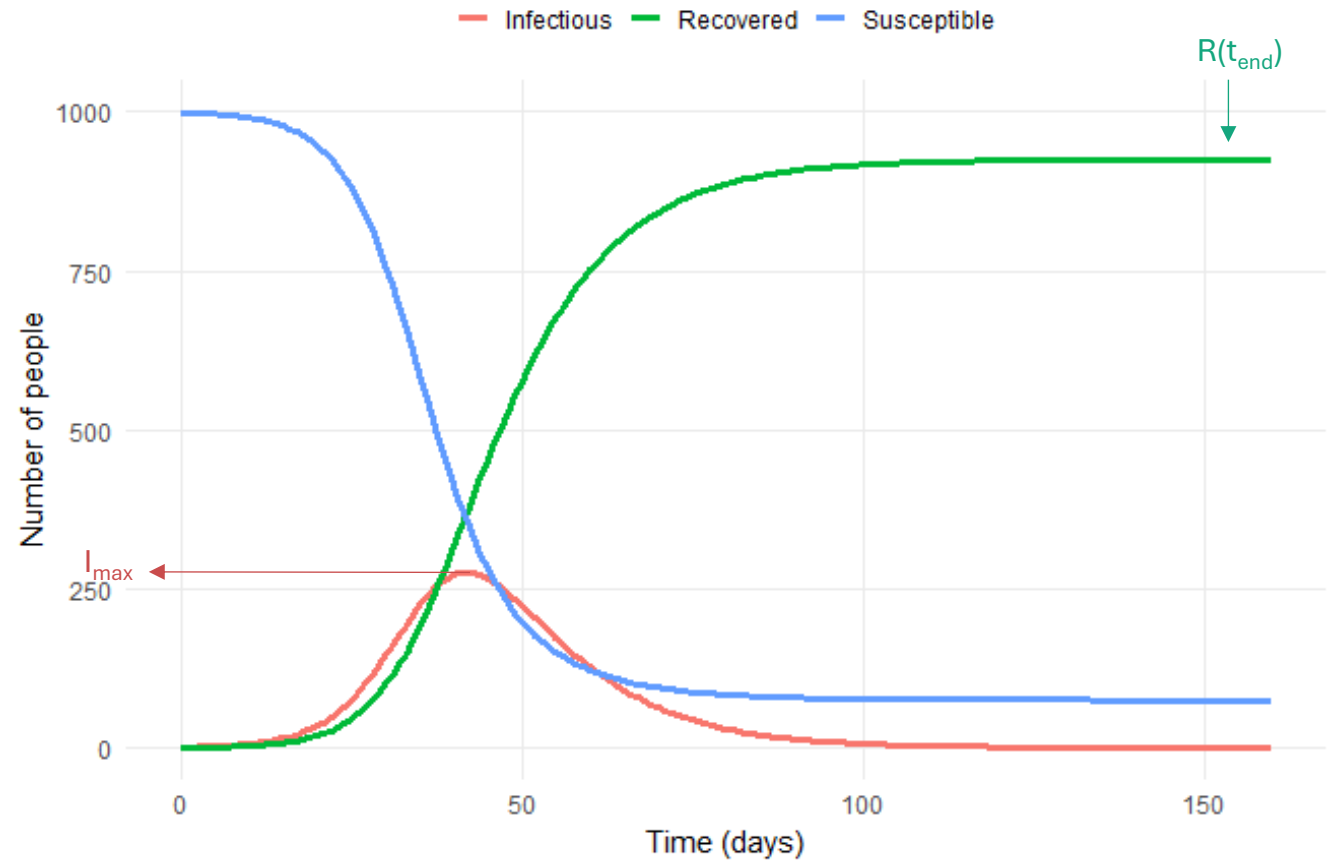
Taken from the epidemic curves



What is the maximum number of infected people at a time? $I_{\max}$?

How many will have gone through an infection?

$$R(t_{\text{end}}) = -S(t_{\text{end}}) + I_0 + S_0$$



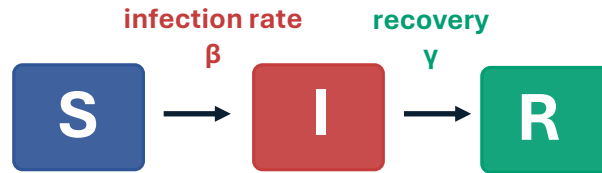
R_0 - potential for transmission

Describes the pathogen–population starting point of a disease

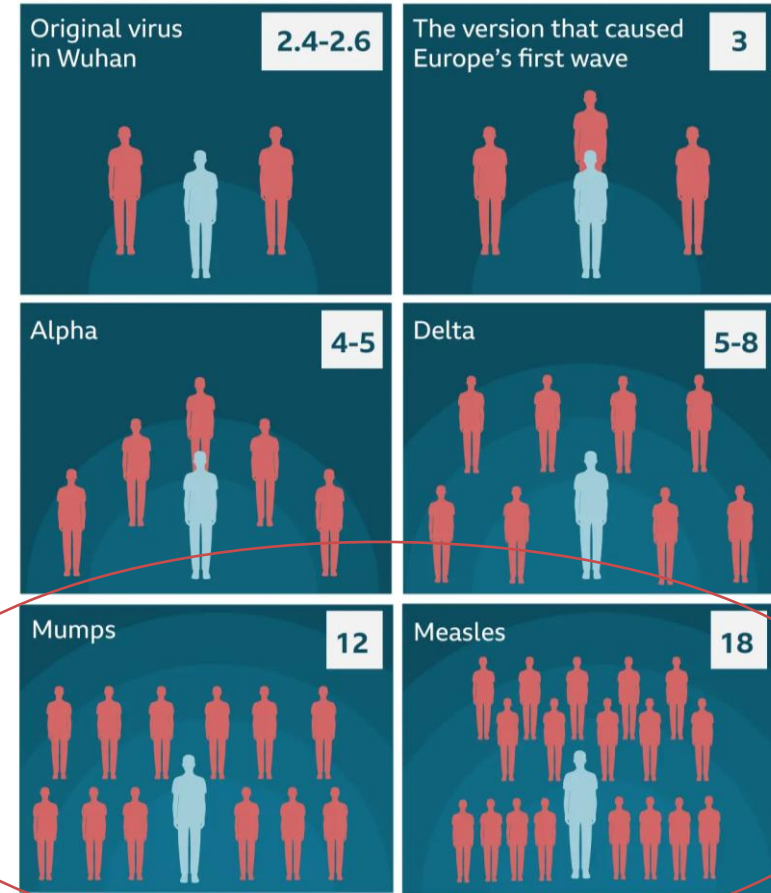
Basic reproduction number

$$R_0 = \beta / \gamma$$

secondary infections from
1 infectious person
in a fully susceptible population
& homogeneous mixing



Common flu between 0.9 and 2.1
depending on the variants circulating



Most disease are already around for long and we have to back calculate
from the behavior we see what we expect R_0 to be.

Source: Imperial College, Lancet, Australian government

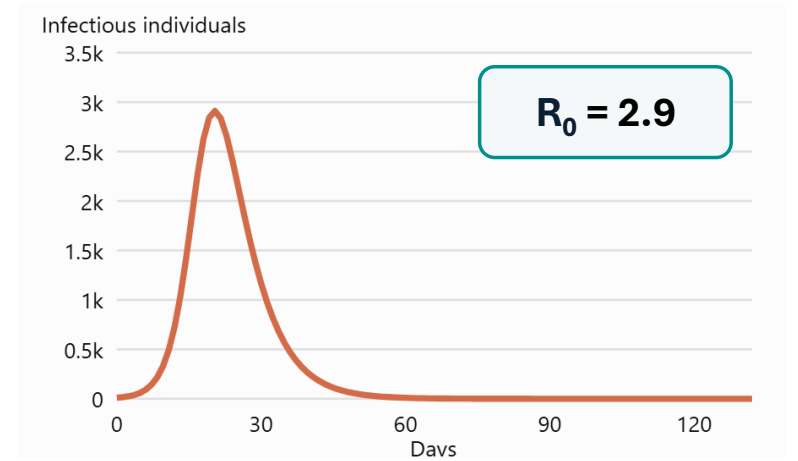
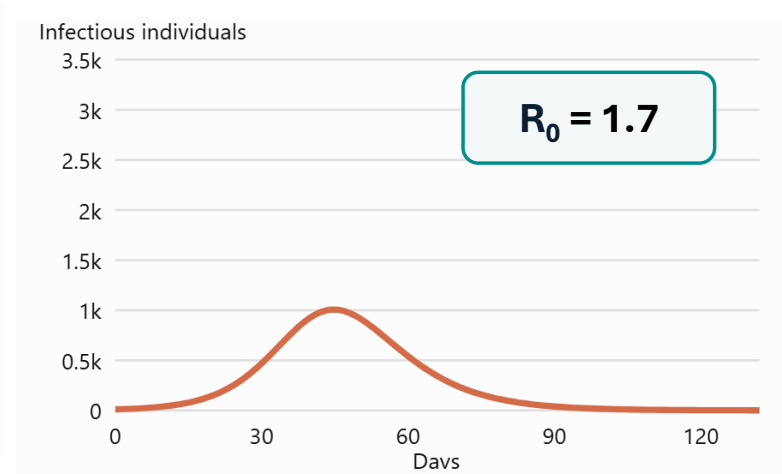
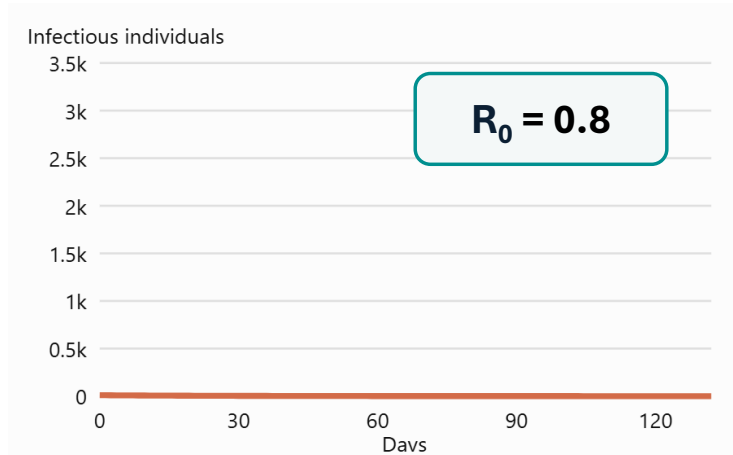
BBC

Will the infection spread – the epidemic take off

Re answers one of the main epidemiological questions

With a new pathogen at the
very start of an epidemic:

$R_0 \geq 1 \rightarrow$ infections don't take off
 $R_0 < 1 \rightarrow$ infections decline



Will the current epidemic worsen, or cases soon drop?

$R_e(t)$ current transmission under current immunity, behaviors and interventions

Basic reproduction number

$$R_0 = \beta / \gamma$$

secondary infections from
1 infectious person
in a fully susceptible population
& homogeneous mixing

Effective reproduction number

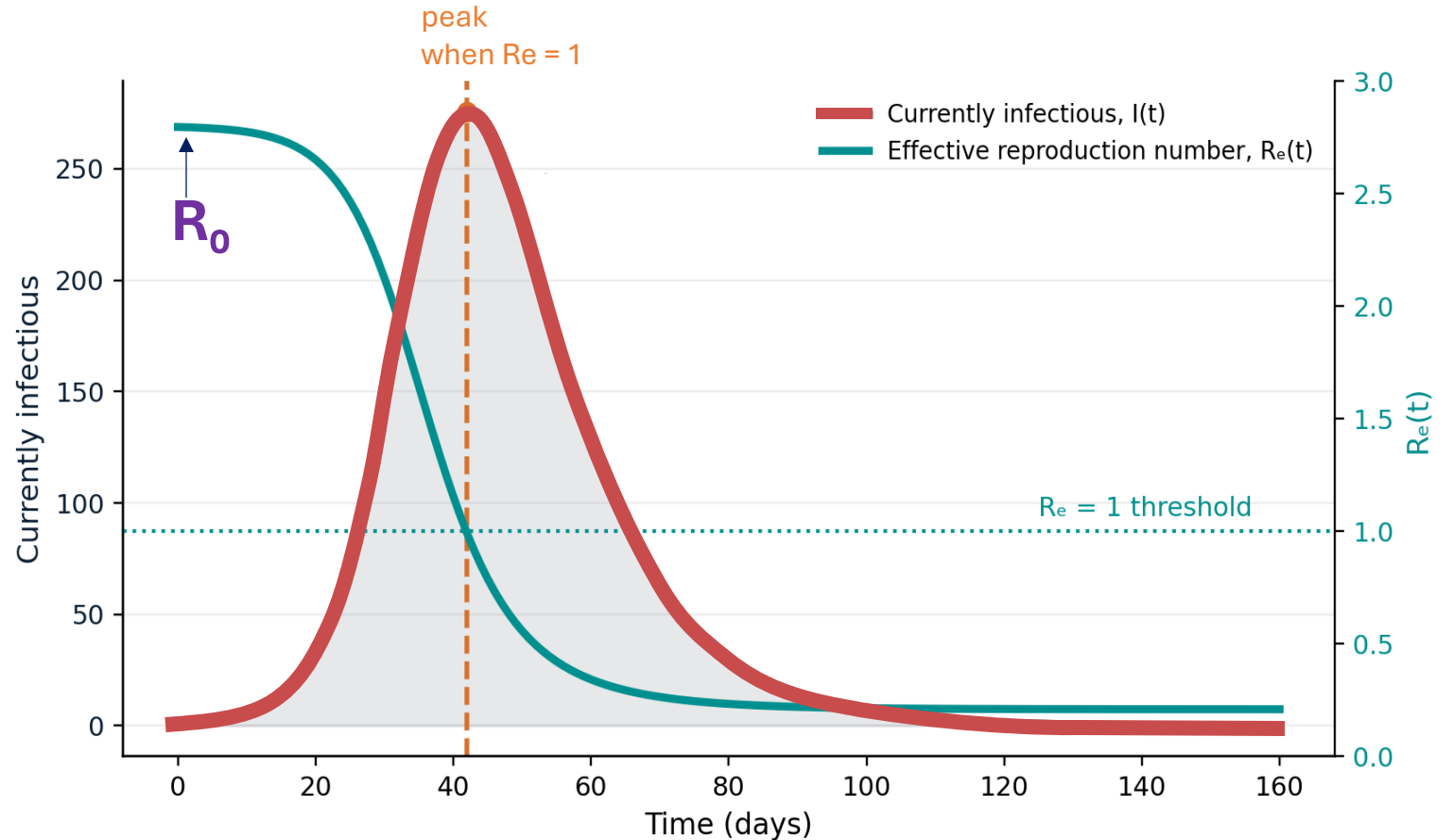
$$R_e(t) = R_0 \times S(t)/N$$

current transmission after immunity,
depletion and control measures

$R_e > 1 \rightarrow$ infections increase

$R_e = 1 \rightarrow$ epidemic peak

$R_e < 1 \rightarrow$ infections decline



R_e changes over time as susceptible are depleted or interventions reduce transmission.

Usefulness of forward simulations

“what happens if these assumptions are true?” Parameter assumption/scenario sensitivity analysis



Two common simulation modes

Deterministic

Same inputs → same curve

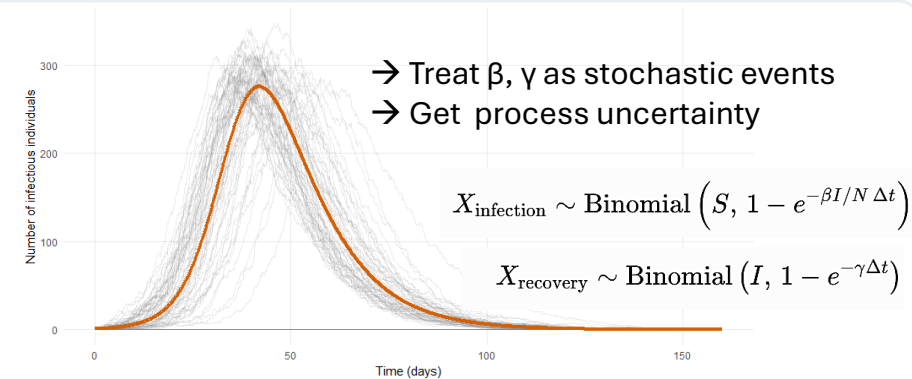
Useful for transparent first principles; quick & simple

Stochastic

Same inputs → many possible outbreaks

Useful when chance matters

→ e.g. in emergence, where the 1st infectious person might not transmit to anyone



Simulation is forward reasoning → It does not “learn” β or γ from data
→ it shows the consequences of values you provide.

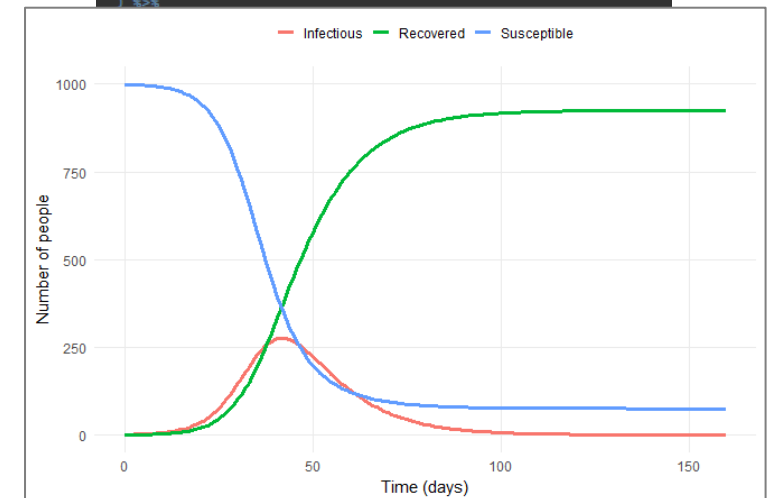
How does this look in R

Example for an implementation of a deterministic SIR model

```
# =====  
# 1. BASIC SIR MODEL  
# =====  
  
sir ← function(t, y, parms) {  
  
  S ← y["S"]  
  I ← y["I"]  
  R ← y["R"]  
  
  beta ← parms["beta"]  
  gamma ← parms["gamma"]  
  
  N ← S + I + R  
  
  # Flow from susceptible → infectious  
  incidence ← beta * S * I / N  
  
  dS ← -incidence  
  dI ← incidence - gamma * I  
  dR ← gamma * I  
  
  list(  
    c(  
      dS,  
      dI,  
      dR  
    ),  
    incidence = incidence  
  )  
}
```

```
# =====  
# 2. HELPER FUNCTION TO RUN A MODEL  
# =====  
  
simulate_sir ← function(  
  beta,  
  gamma,  
  S0 = 999,  
  I0 = 1,  
  R_start = 0,  
  days = 160,  
  scenario = "Scenario"  
) {  
  
  y0 ← c(  
    S = S0,  
    I = I0,  
    R = R_start  
  )  
  
  parms ← c(  
    beta = beta,  
    gamma = gamma  
  )  
  
  out ←  
    ode(  
      y = y0,  
      times = 0:days,  
      func = sir,  
      parms = parms  
    ) %>%  
    as.data.frame() %>%  
    mutate(  
      scenario = scenario,  
      beta = beta,  
      gamma = gamma,  
      R0_basic = beta / gamma,  
      infectious_period = 1 / gamma,  
      N = S + I + R,  
      Re = R0_basic * S / N  
    )  
  
  return(out)  
}
```

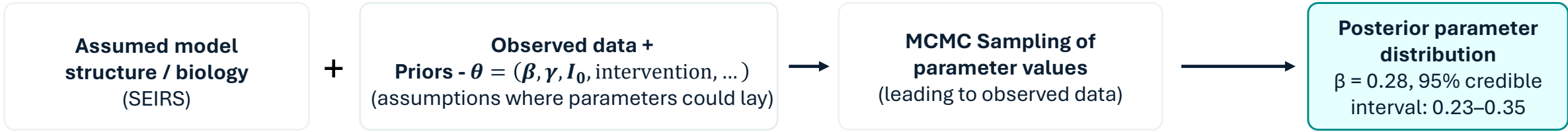
```
# =====  
# 3. BASELINE EPIDEMIC  
# =====  
# beta = 0.28 / day  
# gamma = 0.10 / day  
#  
# Mean infectious period ~ 10 days  
# R0 = 2.8  
# =====  
  
baseline ←  
  simulate_sir(  
    beta = 0.28,  
    gamma = 0.10,  
    scenario = "Baseline: R0 = 2.8"  
  )  
  
# PLOT 1  
# CLASSIC SIR DYNAMICS  
# =====  
  
baseline_long ←  
  baseline %>%  
  select(  
    time,  
    Susceptible = S,  
    Infectious = I,  
    Recovered = R  
  ) %>%
```



```
,  
x = "Time (days)",  
y = "Number of people",  
colour = NULL  
) +  
  
theme_minimal(base_size = 15) +  
  
theme(  
  legend.position = "top",  
  panel.grid.minor = element_blank()  
)  
p1
```

Data & Markov Chain Monte Carlo sampling for parameter search

Fitting plausible parameter values to observed data



Bayes Theorem

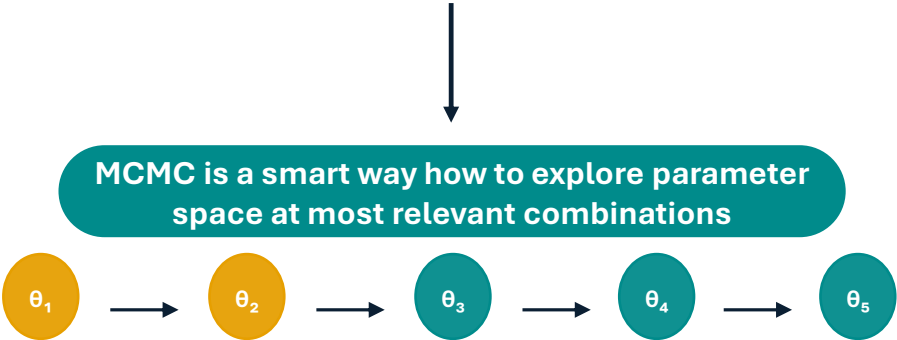
Posterior Likelihood Prior

$$P(\theta \mid \text{data}) \propto P(\text{data} \mid \theta) * P(\theta)$$

$P(\theta)$: A parameter combination can be less or more plausible under our prior

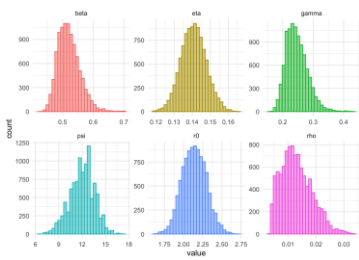
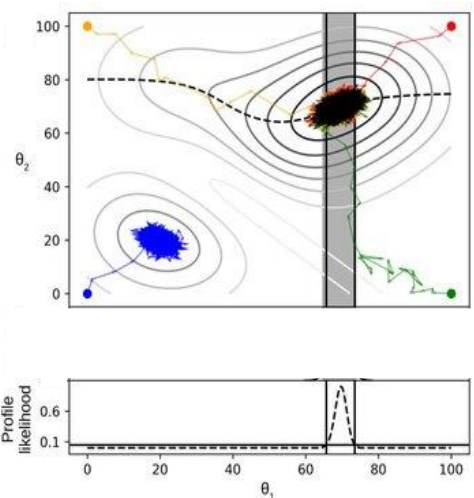
Likelihood/ $P(\text{data} \mid \theta)$: If these parameter values were true, how plausible would the observed data be?

→ We optimize for both at the same time



Propose →
simulate/evaluate
likelihood
→ accept/reject →
repeat many times

Data needs to be
good enough

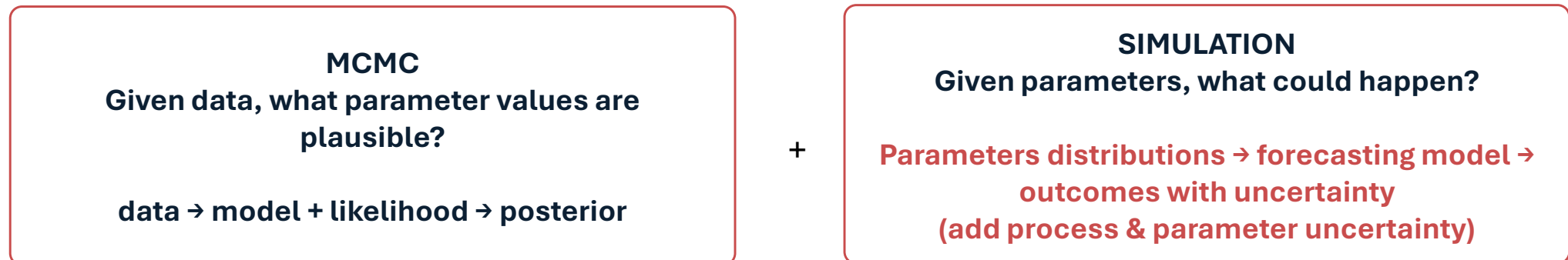
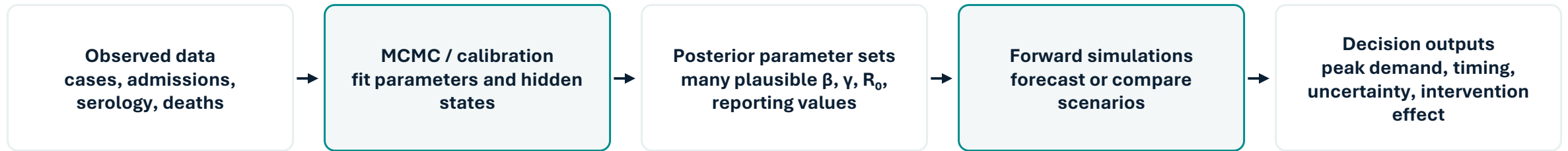


Not just one value

MCMC papers: Gilks et al. 1996; Gelman et al. 2013; Keeling & Rohani 2008.
Picture ref: Villaverde AF, A protocol for dynamic model calibration (2022) *Briefings in Bioinformatics*, A protocol for dynamic model calibration.

How they work together in infectious disease modelling

Recap



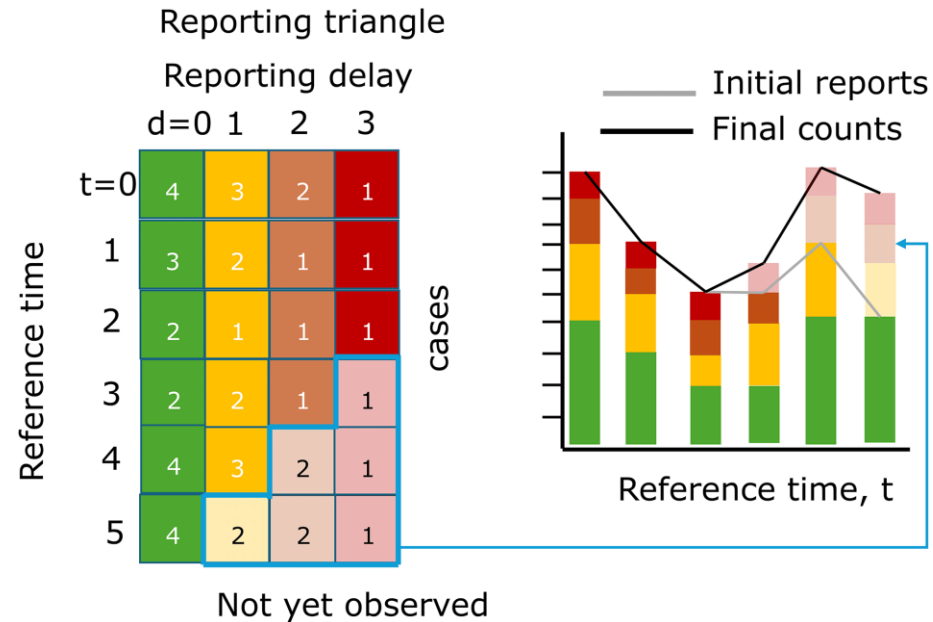
Case data – But what is actually happening right now?

Nowcasting corrects this right-edge bias of incomplete case data - important for situational awareness and short-term planning

reported today != infections today

adding a typical delay distribution
 $P(\text{report delay})$ from past delays

to adjust current day case counts



infection /
onset

care seeking /
test

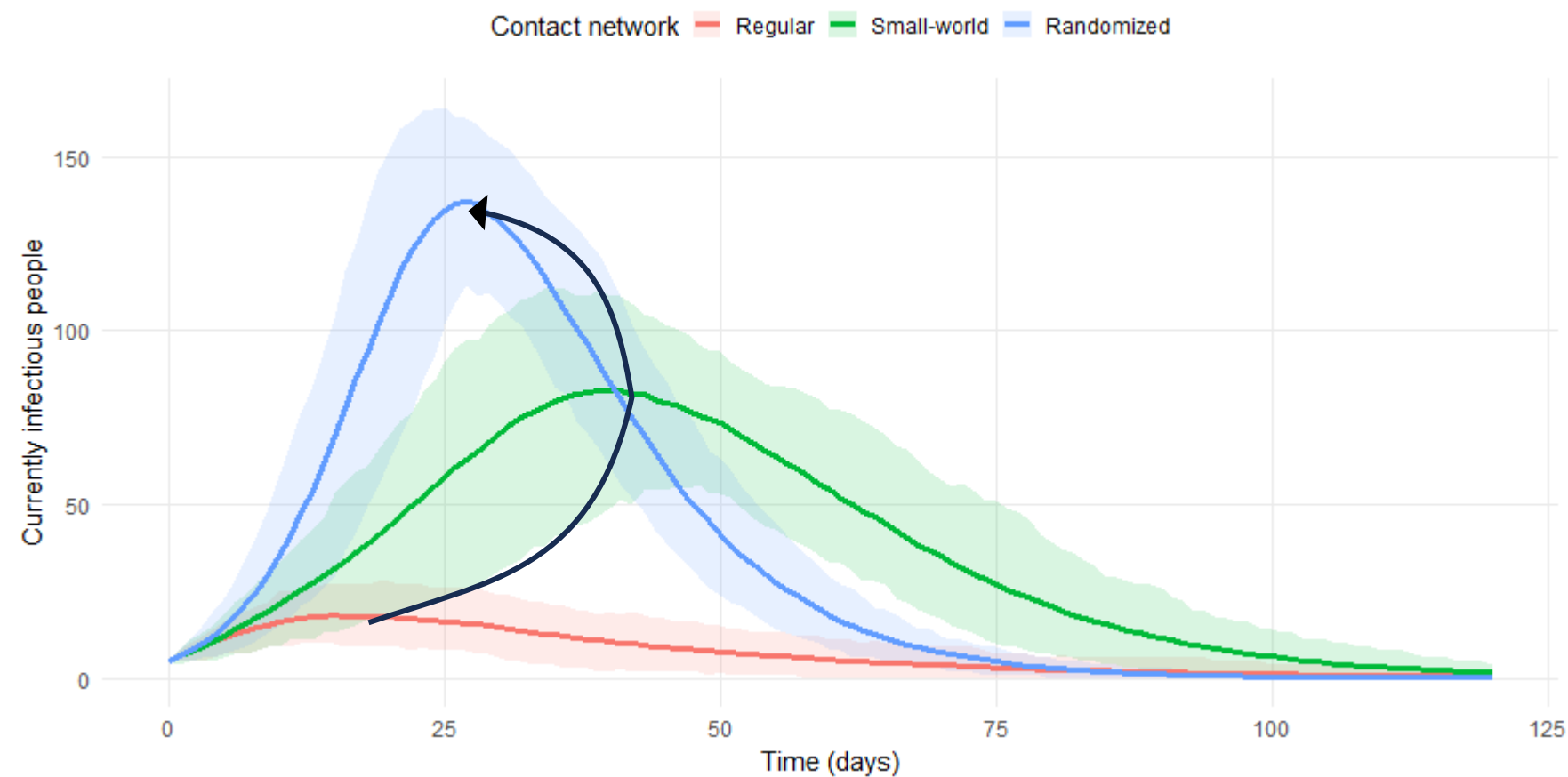
laboratory /
report

Authorities
collection/reporting lag

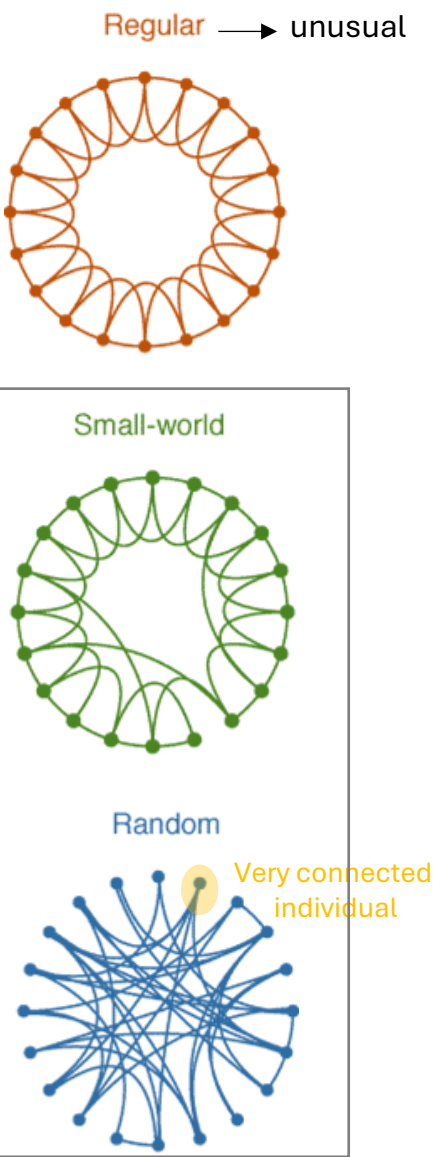
nowcasting != forecasting → estimates the present from incomplete past reports.

Contact networks and mixing matters

Variation leads to different $\beta(t)$

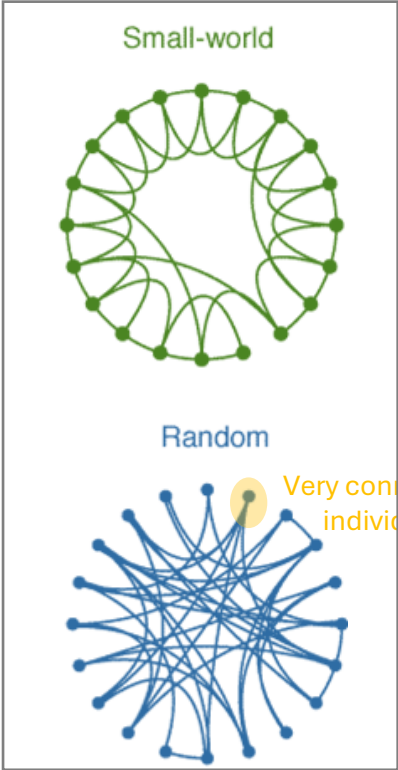
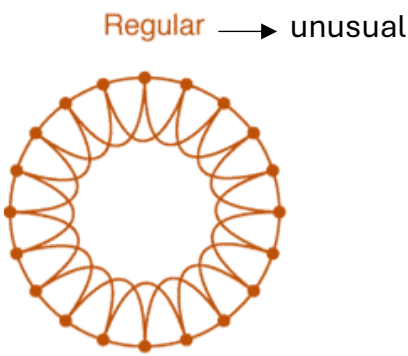
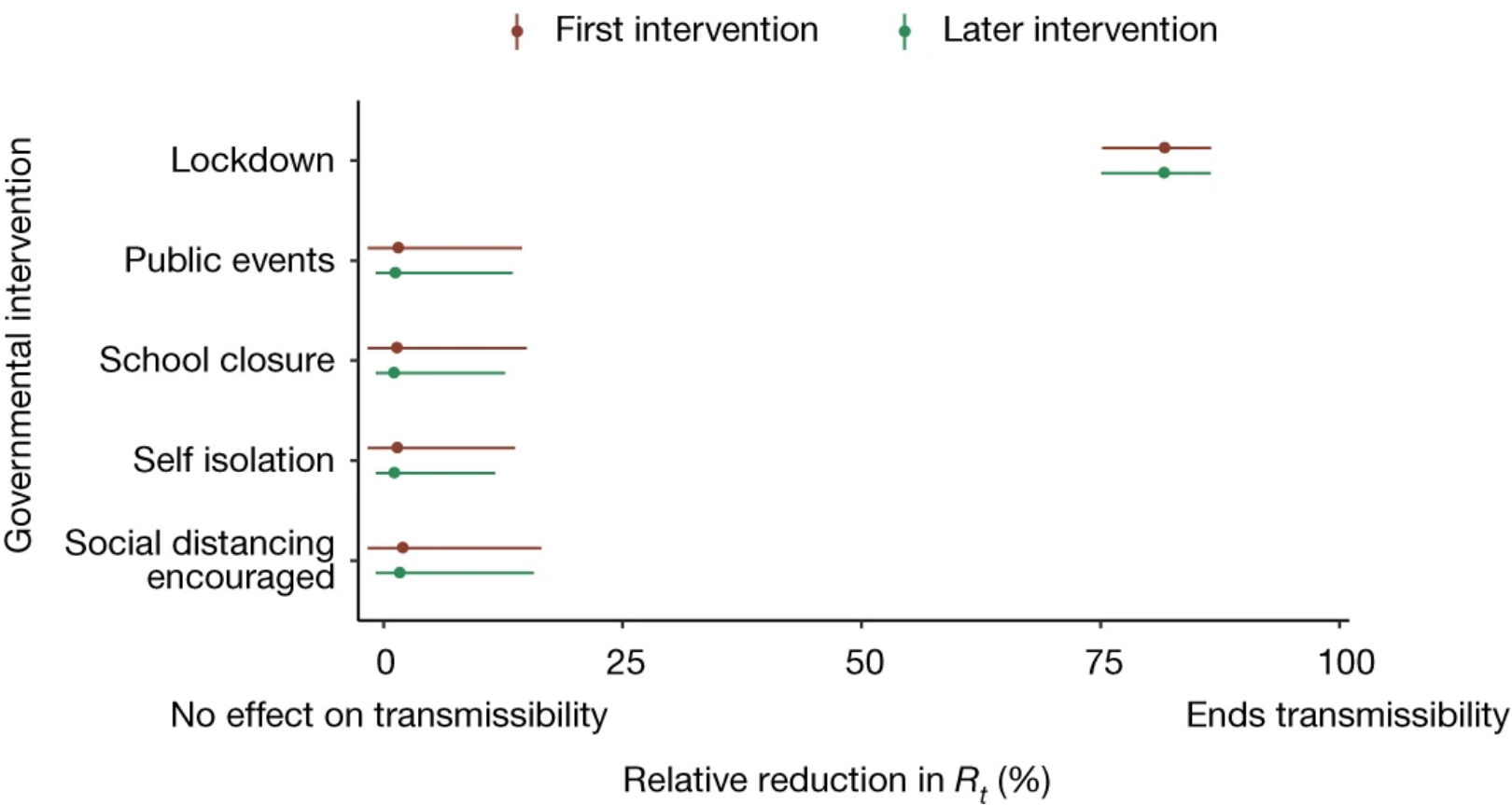


→ Impact of social distancing interventions



Comparing different social distancing measures

Variation leads to different $\beta(t)$

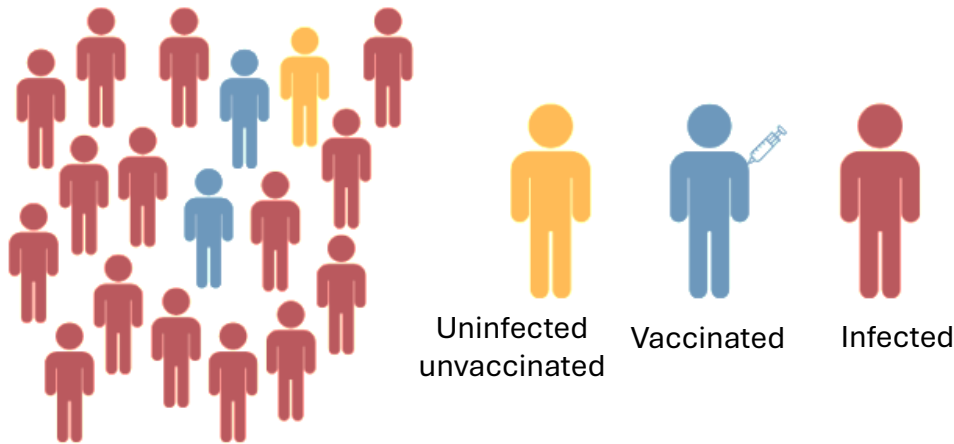
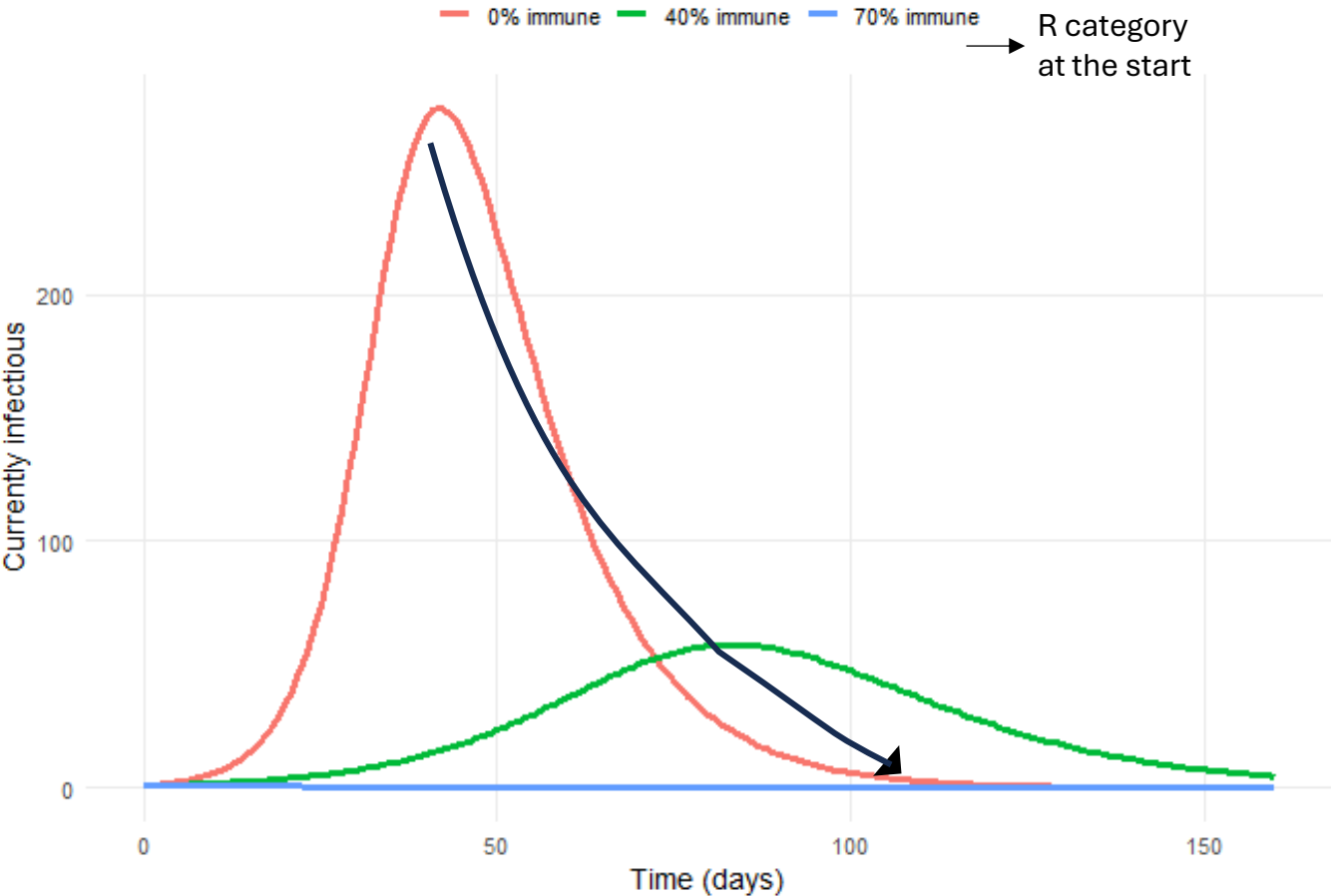


Flaxman et al (2020) *Nature*. Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe
Other example: Perm et al (2020) *Lancet Public Health*. The effect of control strategies to reduce social mixing on outcomes of the COVID-19 epidemic in Wuhan, China: a modelling study

Importance of Immunity/vaccination

Herd immunity protects everyone

The pathogen has the same $R_0 = 2.8$ in every scenario.



Without/few vaccinated people
→ Infections spreads widely

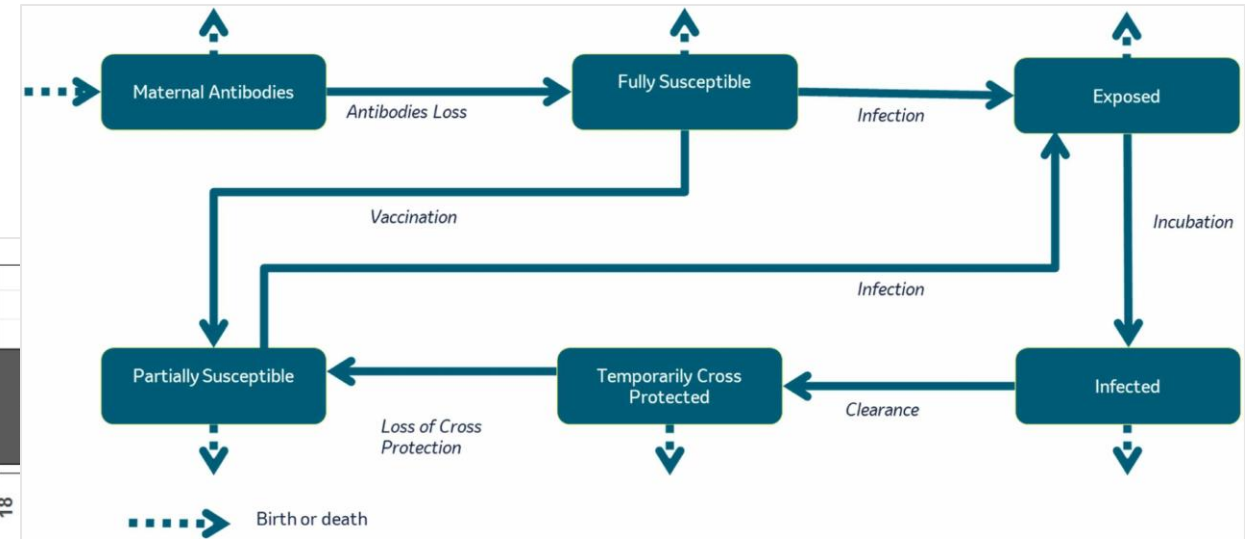
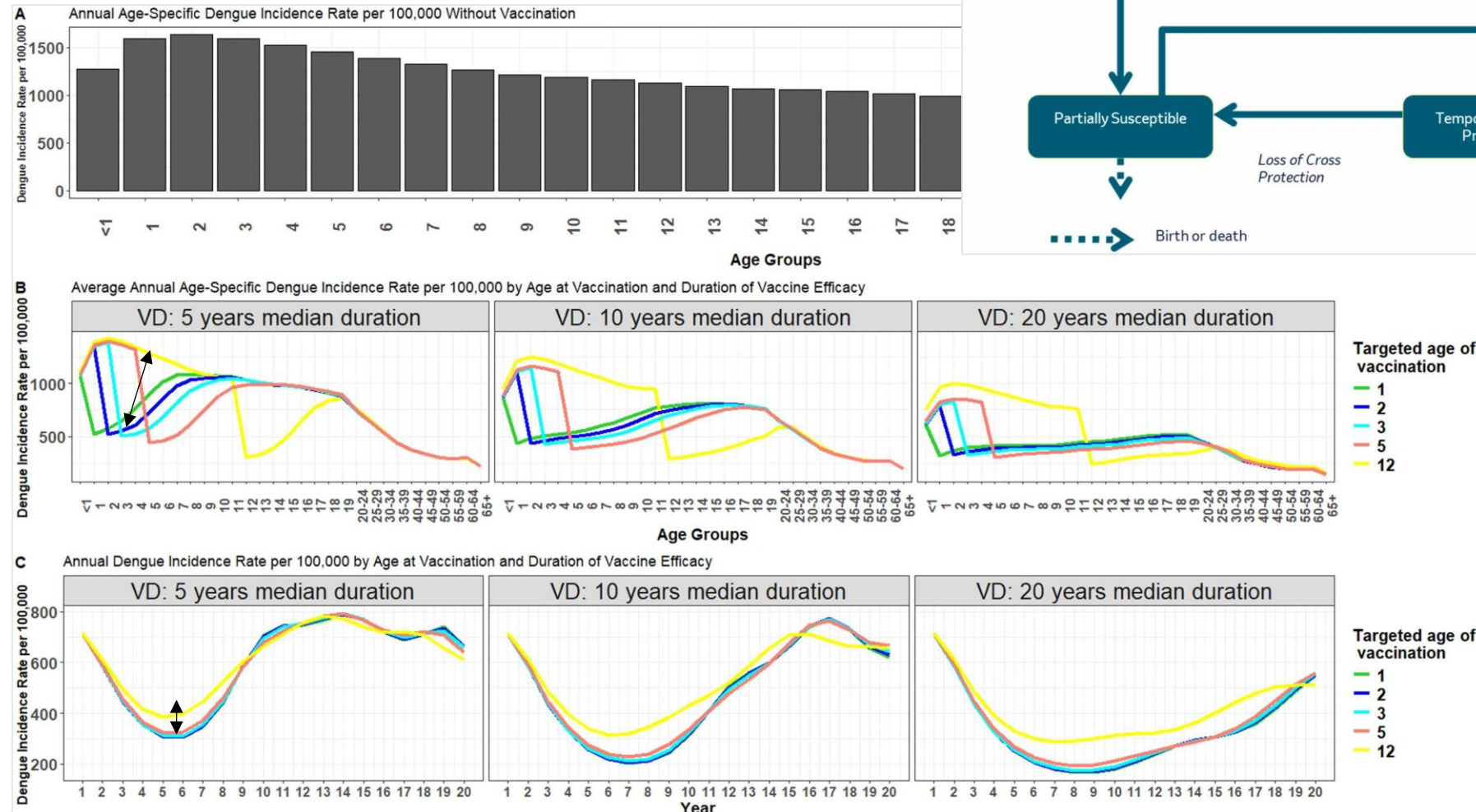
Important parameters:
Past infections/Seroprevalence,
immunity conveyed by vaccination



Higher vaccination rate
sources of infection get isolated

Vaccination roll out

Scenario modelling accounting for age structure and complex immunity stages



Hypothetical Dengue vaccine model for Indonesia

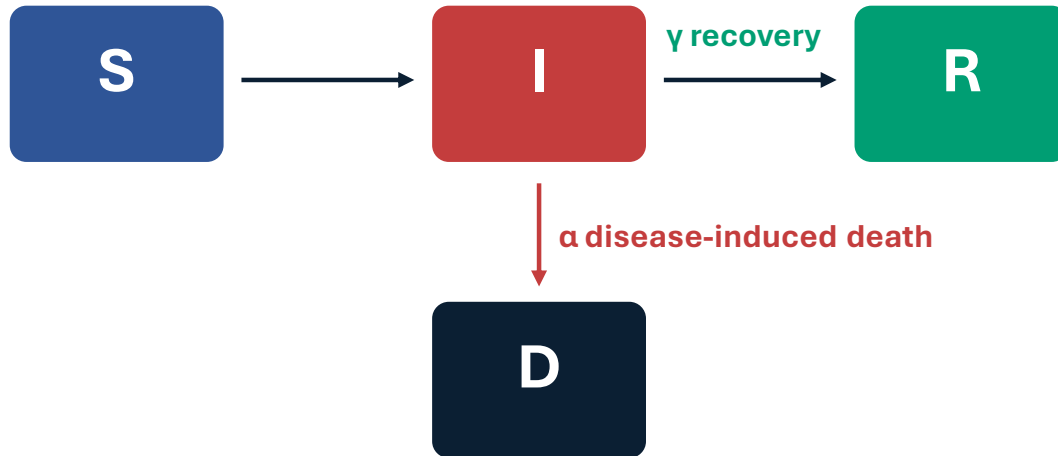
→ **equally safe across ages and serostatus groups** (not the case for Dengvaxia or Qdenga)

→ **routine vaccination** at approximately 1–3 years of age greatest population-level reduction of dengue burden

Importance of Virulence/severity

Death is one of the indicators for disease burden

higher virulence - does not automatically mean - higher spread



$$\begin{aligned} \frac{dI}{dt} &= \beta SI/N - \gamma I - \alpha I \\ \frac{dD}{dt} &= \alpha I \end{aligned}$$

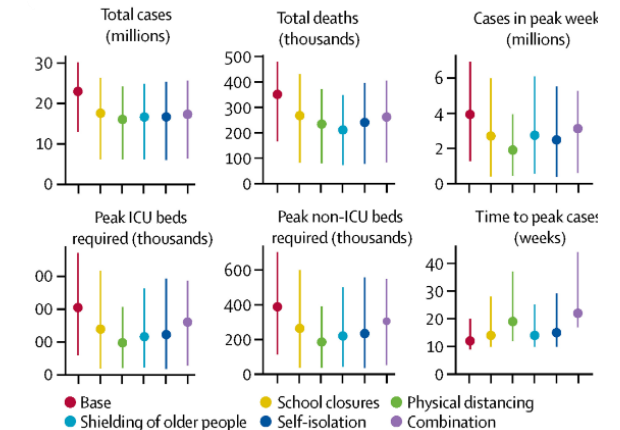
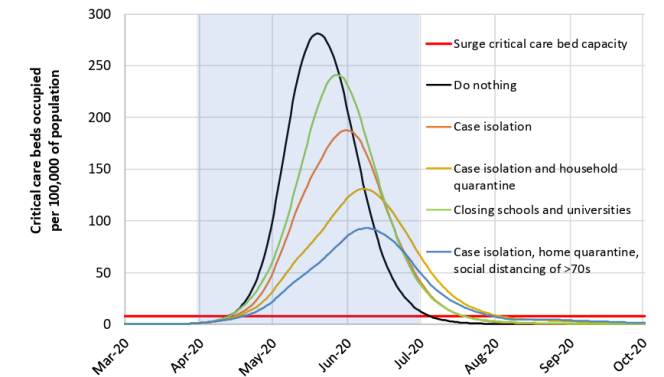
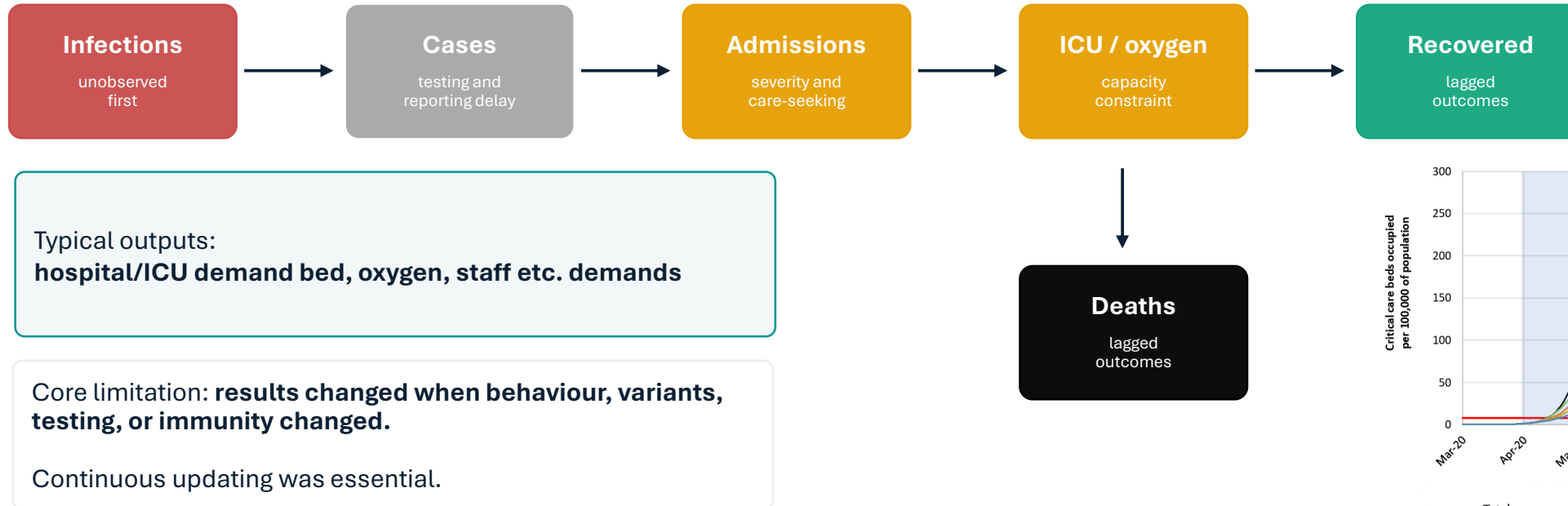
$$R_0 = \beta / (\gamma + \alpha)$$

if virulence shortens infectious duration

Planning insight:
severity parameters drive demand for care (hospital beds, oxygen, ICU),
while transmission parameters drive case numbers.
→ Both are needed for service planning.

Modelling and COVID-19 public health planning

The strongest use was not exact prediction, but stress-testing timing, capacity and intervention scenarios



Real time examples:

Ferguson on behalf of Imperial College COVID-19 Response Team (2020) Report 9: Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand.

Davies on behalf of LSHTM Centre for the Mathematical Modelling of Infectious Diseases COVID-19 working group (2020) *Lancet Public Health*. Effects of non-pharmaceutical interventions on COVID-19 cases, deaths, and demand for hospital services in the UK: a modelling study

What data is needed for what type of question?

The data requirement depends on whether you estimate, explain or plan

Question	Minimum useful data	Model output
Will an outbreak take off? Worsen?	Initial S, contact/transmission rates, infectious duration	Threshold $R_0 > 1$, Threshold $R_e > 1$
Will cases rise next week?	Recent incidence + reporting delay	Short-term nowcasting: # cases now
How to design a vaccination program?	Seroprevalence, vaccination efficacy, previous infection, waning of immunity	target age/risk groups, coverage, timing, and predicted cases averted
Which intervention works best?	Mechanism of action + plausible ranges + uptake/adherence	Scenario (case/burden) comparison
How many beds are needed?	Incidence, severity, admission risk, length of stay, delays	peak occupancy, timing, and number of beds required

Uncertainty is the main output decision-makers need: → best case, plausible range, worst case.

Reality check

Barriers to infectious-disease modelling informing health-care planning in practice

Data gaps → some data very costly/incomparable

under-reporting, delayed reporting, changing of testing

Parameter uncertainty

Constant evolution of pathogens

Wrong structure

missing age, space, networks, vectors, care pathways

Behavioural feedback

people and policies change when risk changes

Decision mismatch

model outputs may not align with operational decisions

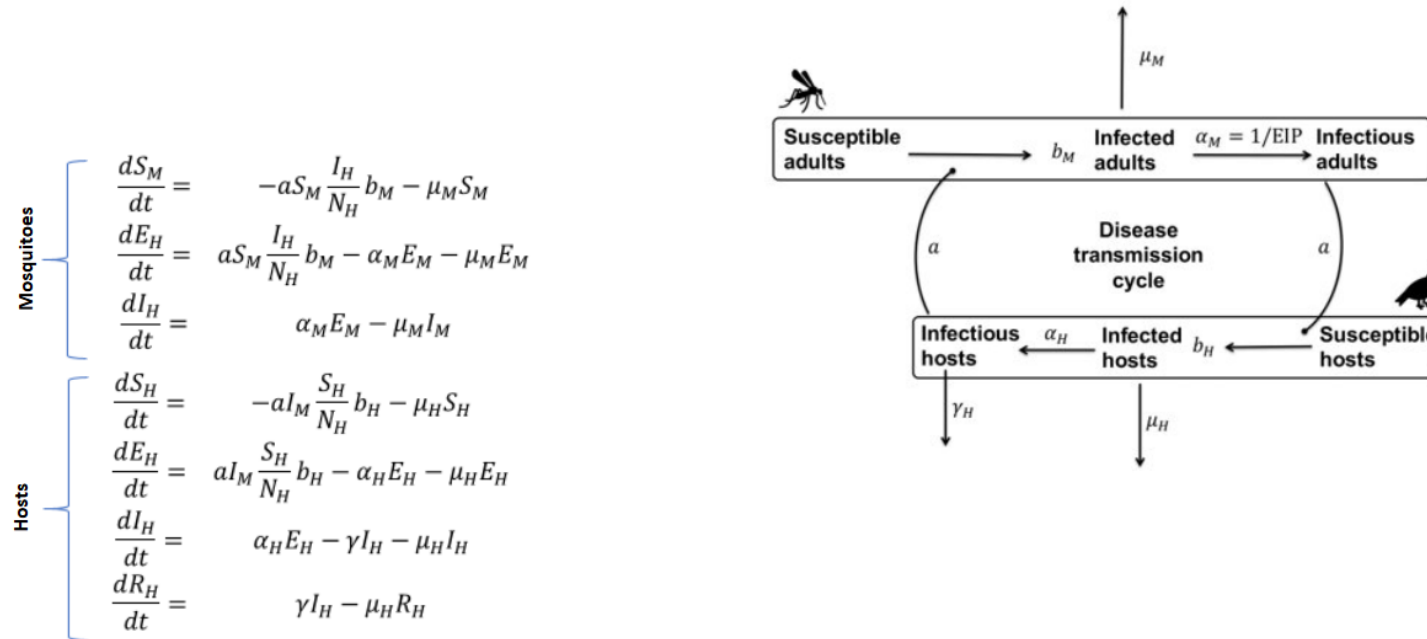
Communication

Model outputs can be misunderstood, or miscommunicated etc.

→ more about intuition and directionality

Heidelberg Planetary Health Hub

Complex interactions between the environment and infectious disease – a lot of mosquito born disease modelling



$$R_0 = \frac{ma^2 b_M b_H \frac{\alpha_M}{\alpha_M + \mu_M} \frac{\alpha_H}{\alpha_H + \mu_H}}{\mu_M (\gamma_H + \mu_H)} = \frac{\alpha_H}{\alpha_H + \mu_H} * \frac{1}{(\gamma_H + \mu_H)} * ma * b_M * \frac{\alpha_M}{\alpha_M + \mu_M} * \frac{1}{\mu_M} * a * b_H$$

$$m = \frac{N_M}{N_H} \quad \text{"Mosquito (M)-to-host (H) ratio"}$$

